

Grid-Integrated Data Centers: Architectures, Modeling, Interconnection Requirements, Challenges, and Opportunities

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ABSTRACT Data centers are emerging as one of the fastest-growing electricity consumers worldwide due to the rapid expansion of cloud computing, artificial intelligence (AI), and digital services. The large-scale integration of AI data centers into electric power systems introduces significant challenges for grid planning, operation, stability, power quality, and compliance with evolving grid codes. Modern data centers are characterized by power-electronic-dominated infrastructures, including high-density computing platforms, advanced cooling systems, on-site renewable energy, and energy storage resources, all of which exhibit dynamic behaviors distinct from conventional passive loads. Consequently, their increasing integration necessitates the development and application of appropriate modeling frameworks to accurately assess grid impacts and enable effective control and coordination strategies. This article covers grid-integrated data centers, with a focus on their electrical and cooling architectures, and associated modeling approaches. In addition, modeling of critical data center components, interconnection requirements, and key integration challenges is examined. Finally, emerging opportunities for data centers to provide frequency regulation and flexibility services are discussed, outlining future research directions toward reliable, efficient, and grid-interactive data center integration.

INDEX TERMS Artificial intelligence (AI) data center, data center modeling, fault ride-through (FRT), graphics processing unit (GPU), hybrid cooling, power quality, power system, uninterruptible power supply (UPS).

NOMENCLATURE

AESO	Alberta Electric System Operator.	FRT	Fault ride-through.
AI	Artificial intelligence.	GFM	Grid-forming.
BESS	Battery energy storage system.	GPU	Graphics processing unit.
CDU	Coolant distribution unit.	HIL	Hardware in the loop.
CPU	Central processing unit.	HVAC	Heating, ventilation, and air conditioning.
CRAC	Computer room air conditioner.	ICS	Industrial control systems.
CRAH	Computer room air handler.	LV	Low voltage.
CRU	Commission for regulation of utilities.	LVRT	Low-voltage ride-through.
EMT	Electromagnetic transient.	MV	Medium voltage.
ERCOT	Electric Reliability Council of Texas.	NERC	North American electric reliability corporation.
ESS	Energy storage system.	PCC	Point of common coupling.
E-STATCOM	Energy storage static synchronous compensator.	PDU	Power distribution unit.
		PFC	Power factor correction.

PEM	Proton exchange membrane.
POI	Point of interconnection.
PSU	Power supply unit.
PUE	Power usage effectiveness.
SCADA	Supervisory control and data acquisition.
SLA	Service-level agreement.
TPU	Tensor processing unit.
UPS	Uninterruptible power supply.
VASP	Vienna Ab initio simulation package.
VRM	Voltage regulator module.
WECC	Western electricity coordinating council.

I. INTRODUCTION

The unprecedented expansion of digital economy has accelerated the adoption of AI across a wide range of applications, including computer vision, recommendation systems, scientific computing, optimization, and natural language processing. Many of these applications rely on large-scale neural-network-based deep learning algorithms that require extensive training and inference workloads. Generative AI models, such as large language models, including GPT (OpenAI), Gemini (Google DeepMind), and LLaMA (Meta) [1], which are designed to generate human-like text and power advanced conversational agents, drive this growth [2]. The growing reliance on deep learning workloads, characterized by long training cycles and massive parallel processing requirements, is driving rapid expansion of hyperscale computing infrastructure. As a result of this, data centers are scaling in both size and density, leading to a significant and sustained increase in their electricity demand. This surge is placing significant stress on electrical grids, making the integration and sustainable operation of data centers a critical challenge for modern energy infrastructure [1], [3], especially as the power industry simultaneously undergoes large-scale system modernization, decarbonization, and digitalization of its assets [4]. Based on equipment-specific statistics shown in Fig. 1, the International Energy Agency reported that global data centers consumed approximately 415 TWh of electricity in 2024 [5], with conventional servers accounting for the largest share at 46.3%. Cooling systems contributed 18.1%, followed by accelerated servers at 14.5%, with other infrastructure and IT equipment accounting for 10.8% and 10.3%, respectively. This energy consumption is projected to increase significantly, potentially doubling to around 945 TWh by 2030, mainly driven by the expanding computational demands of AI data centers.

Over the past decade, countries with low-carbon electricity have struggled to meet the growing electricity demands of data centers. For instance, in the United States, data centers consumed approximately 183 TWh of electricity in 2024, representing more than 4% of the country's total electricity usage, comparable to Pakistan's annual electricity demand [6]. This consumption is projected to increase to around 260 TWh by 2026, which would represent approximately 6% of the total electricity demand [7]. In Australia, data centers are estimated to have consumed 3.9 TWh in 2025 and is projected to increase to about 12.0 TWh by 2030. Comparable

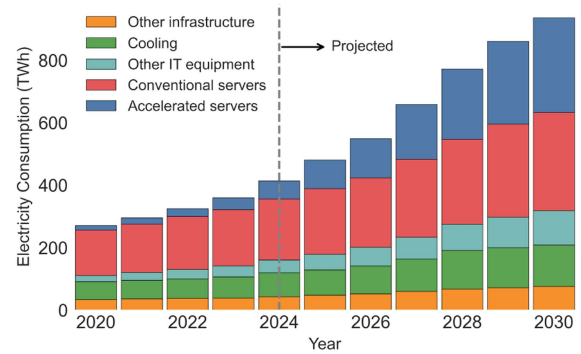


FIGURE 1. Global data center electricity consumption, by equipment, Base Case, 2020–2030.

trends are being observed globally, in countries such as China, Singapore, Ireland, and the United Kingdom [8]. This trend has intensified interest in the concept of grid-integrated data centers and the manner in which their electrical and thermal infrastructures are analyzed within the context of power system operation, planning, and energy management. Fig. 2 illustrates the evolution of these infrastructures from legacy facilities of the early 2000s to traditional large-scale AI data centers, characterized by significant increase in facility size, rack power density, cooling complexity, and electrical architecture advancement.

As the scale and energy intensity of these facilities continued to grow, concerns about their impact on the utility grid prompted system operators to introduce new regulatory and policy measures to maintain grid reliability and resilience. In 2022, data centers significantly increased Ireland's national electricity demand, leading the Commission for Regulation of Utilities (CRU) to restrict new connections in Dublin due to grid capacity limitations [9]. Similarly, the Netherlands suspended new data center permits due to energy use concerns and infrastructure impacts, while Singapore introduced stricter approval frameworks to ensure that future developments align with sustainability goals [10], [11], [12]. Recently, in Canada, the AESO published new guidelines for grid-integrated data centers. These guidelines aim to ensure that data centers are designed, modeled, integrated, and operated, in a manner that maintains grid reliability, stability, and power quality while also addressing operational and planning challenges [13]. In the same way, the Australian Energy Market Commission has also proposed technical standards for large-scale data centers connecting to the National Electricity Market [14].

These measures reflect a growing recognition that data center expansion can pose significant risks to grid reliability. Moreover, as regulatory frameworks and grid support expectations continue to evolve, modern data centers are increasingly transitioning from passive energy consumers to grid-interactive infrastructures. In this context, grid-interactive data centers actively respond to grid conditions through coordinated mechanisms such as demand response, computational workload shifting, renewable energy integration, battery

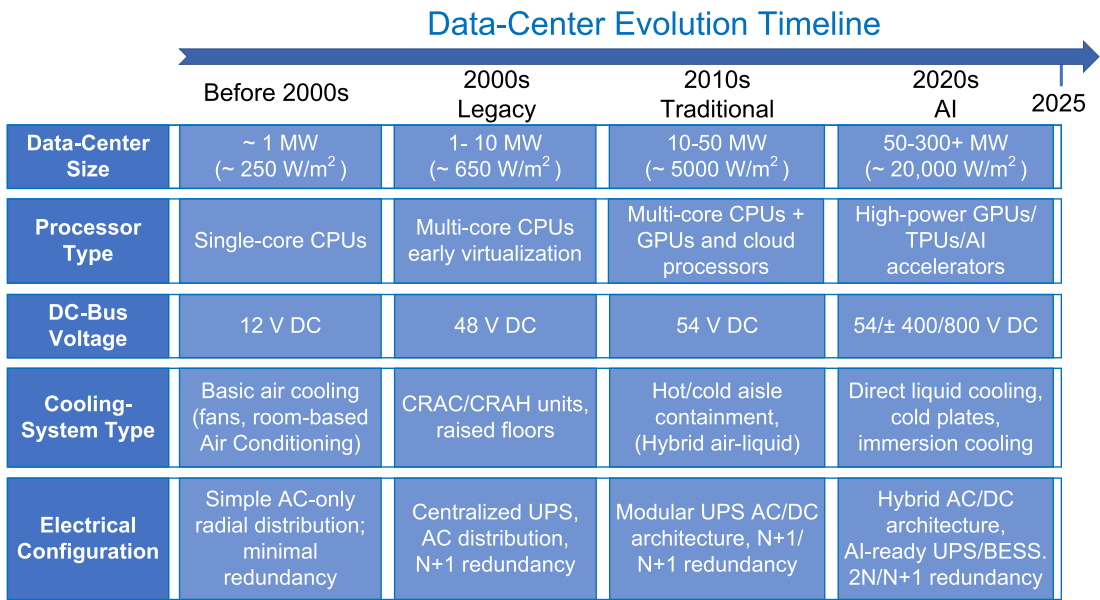


FIGURE 2. Evolution of data center characteristics across four generations.

TABLE 1. Comparison of Existing Literature and Scope Coverage of This Work

Paper	Year	Architectures		Modeling	Interconnection Requirements	Challenges	Opportunities
		Electrical	Cooling System				
[15]	2015	✓	✓	✓	×	×	×
[1]	2025	×	×	×	×	✓	✓
[19]	2025	×	×	×	×	✓	×
[17]	2025	×	✓	×	×	✓	×
[16]	2025	×	✓	×	×	✓	✓
[18]	2026	×	×	×	×	✓	✓
This work		✓	✓	✓	✓	✓	✓

energy storage dispatch, backup generation scheduling, and participation in ancillary service markets. This transformation positions data centers not only as major electricity consumers but also as flexible strategic assets capable of supporting power system stability, resilience, and decarbonization goals.

This transitioning toward grid-interactive operation creates an urgent need for a comprehensive understanding of data center architectures, modeling approaches, interconnection requirements, and grid integration challenges and opportunities. In particular, the increasing coupling between computational workloads, electrical infrastructures, cooling systems, energy storage, and power system dynamics necessitates more advanced and integrated modeling frameworks capable of accurately representing data center behavior under diverse operating conditions. Such understanding is essential not only for researchers and system operators, but also for informing future regulatory and planning frameworks capable of balancing continued digital innovation with power system reliability, resilience, and sustainability.

A. LITERATURE LIMITATIONS

As illustrated in Table 1, a considerable number of review papers have examined grid-integrated data centers, with most

works concentrating on specific aspects such as electrical architectures, cooling systems, modeling approaches, or the associated challenges and opportunities, rather than providing a comprehensive coverage of all these dimensions simultaneously. A review by Oró et al. [15] provided an overview of existing data center infrastructures, energy efficiency measures, and renewable energy integration approaches, while emphasizing the need for dynamic models and metrics for evaluating energy consumption and emerging efficiency strategies. With the growing interest in grid-integrated and AI-driven data centers, subsequent review studies expanded the discussion toward smart grid integration, distributed energy systems, and advanced cooling technologies [16] [17]. More recent high-quality reviews have further examined the rapidly increasing electricity demand of AI data centers and its implications for modern power systems. In particular, the authors of [1] and [18] comprehensively discussed evolving load characteristics, grid impacts, sustainability concerns, operational challenges, and emerging technological solutions, while Ginzburg-Ganz et al. [19] highlighted the unique grid integration challenges associated with hyperscale AI-driven facilities, including their impacts on system planning, stability, and energy infrastructure.

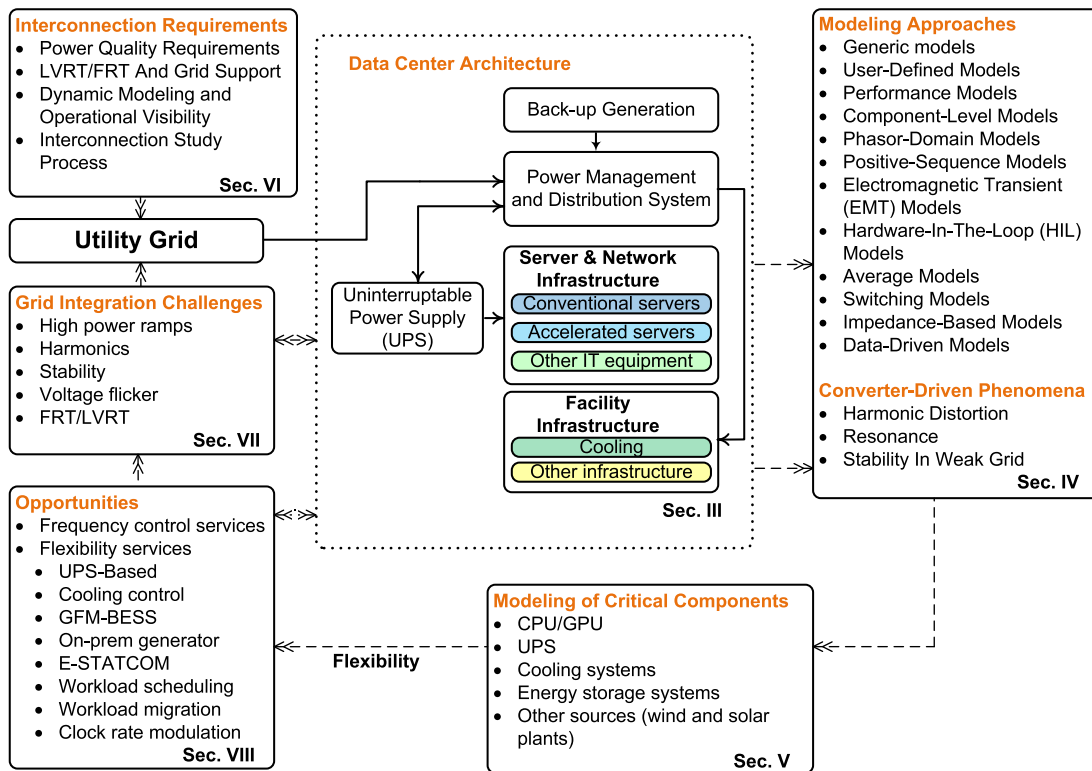


FIGURE 3. Article framework.

Despite the valuable contributions of these studies, limited attention has been given to the combined analysis of electrical and cooling architectures, modeling approaches, interconnection requirements, as well as grid integration challenges and emerging opportunities. This highlights the need for a more comprehensive review framework that systematically examines these closely related aspects within the context of modern grid-integrated and grid-interactive data centers.

B. CONTRIBUTIONS

This article presents a comprehensive and structured review of grid-integrated data centers, with particular focus on their architectures, modeling approaches, and interaction with modern power systems. In contrast to existing review studies, this article examines the evolution of data centers from legacy infrastructures to traditional facilities and, ultimately, to AI data centers, while highlighting the operational transition from grid-integrated to grid-interactive paradigms. The main contributions of this article are summarized as follows.

- 1) This article presents a critical analysis of data center architectures, discussing the evolution of AC, DC, and hybrid electrical distribution systems alongside air-based, liquid-based, and hybrid cooling strategies, with emphasis on their efficiency and suitability for high-density computing workloads.
- 2) It examines 12 modeling approaches for grid-integrated data centers, detailing their underlying assumptions, scope, applicability, advantages, and limitations, and

provides a perspective on converter-driven phenomena, including harmonics, resonance, and stability in weak grids. In addition, it presents a structured model selection framework that connects study objectives and time scales to the required modeling parameters.

- 3) It clarifies the interconnection requirements for large-scale grid-integrated data center loads by distinguishing between mandatory industry practices and recommended ones.
- 4) New insights into modeling critical data center components are provided, focusing on their dynamic interactions and applicability for grid integration and power system stability studies.
- 5) This article examines the challenges and opportunities of integrating modern data centers with the grid.

C. ARTICLE FRAMEWORK AND ORGANIZATION

The framework illustrated in Fig. 3 clearly connects data center architecture, modeling approaches, and grid integration considerations. The physical architecture of a data center forms the structural foundation for developing both device-oriented and system-level models. These models facilitate the analysis of dynamic behaviors, control interactions, and performance characteristics that directly affect grid-related issues. The framework provides a cross-layer perspective by linking architectural elements with relevant modeling paradigms and interconnection requirements, challenges, and opportunities. This perspective is intended to support researchers in the coordinated design, analysis, and operation of

TABLE 2. Keyword Categories Used for Literature Search

Category	Keywords	Purpose
Set 1	"data center", "AI data center", "hyperscale data center", "cloud computing", "high-performance computing", "GPU cluster", "grid-integrated", "grid-interactive"	Define the scope of the target infrastructure sector
Set 2	"electrical architecture", "cooling architecture", "hybrid AC/DC", "liquid cooling", "air cooling", "energy management"	Specify operational and architectural characteristics
Set 3	"data center modeling", "dynamic model", "electrical model", "thermal model", "power system modeling", "energy modeling", "simulation", "load modeling"	Specify modeling and analytical approaches
Set 4	"Grid code", "interconnection standards", "connection requirements", "transmission connection", "reliability standards"	Define interconnection and compliance requirements
Set 5	"load variability", "grid stability", "power ramps", "harmonics", "voltage flicker", "FRT/LVRT", "power quality", "transmission constraints", "cybersecurity", "data privacy"	Capture grid integration challenges
Set 6	"demand response", "grid stability", "renewable integration", "ancillary services", "frequency regulation", "flexibility", "energy storage"	Capture grid integration services and opportunities

data centers, positioning them as active participants in future power systems.

The rest of this article is organized as follows. Section II provides the review methodology. Section III presents an overview of data center architectures. Section IV reviews modeling approaches and converter-driven phenomena applicable to data center for grid studies. Section V then focuses on the modeling of critical components within data center electrical and cooling infrastructures. Section VI discusses interconnection requirements relevant to data center integration, while Section VII examines key integration challenges on the grid. Section VIII highlights the opportunities for data centers to provide flexibility services. Finally, Section IX concludes this article and offers future needs.

II. REVIEW METHODOLOGY

The literature survey is conducted to identify existing studies using several scientific databases, including Google Scholar, IEEE Xplore, IET Digital Library, Scopus, Springer, Wiley, and Taylor & Francis. These databases were chosen to ensure comprehensive coverage of journal and conference papers, technical reports, industry guidelines, and relevant standards related to power systems and data center infrastructures. To ensure systematic coverage of the research domain, a structured keyword-based search strategy was adopted. The search terms are grouped into six thematic categories, as detailed in Table 2. The first keyword category defines the scope of the target infrastructure, followed by the second category, which focuses on the operational and architectural characteristics of data centers. The third category addresses modeling and analytical approaches relevant to grid interaction studies, and the fourth emphasizes interconnection and compliance requirements. The fifth category captures grid integration challenges and their associated operational impacts, whereas the sixth highlights grid support services and flexibility opportunities. The keyword groups were combined using Boolean operators such as "AND" and "OR" to retrieve studies satisfying multiple search criteria simultaneously.

To ensure relevance to contemporary developments, only studies published between 2015 and 2026 were considered. Records that did not align with the review objectives were

excluded, while eligible studies proceeded to a full-text assessment to evaluate their relevance and suitability for inclusion. Following the full-text screening stage, the final set of articles was selected for detailed analysis and categorized according to the key themes of architectures, modeling approaches, interconnection requirements, challenges, and opportunities. In addition, manual reference searching was performed by examining the reference lists of highly relevant papers, particularly review articles, to identify further studies that may not have been captured during the initial database search.

III. DATA CENTER ARCHITECTURES

The architecture of a data center is fundamentally composed of integrated power network, cooling, and IT subsystems designed to ensure continuous, reliable, and efficient operation [20]. Power is supplied from the utility grid through a power management and distribution system that coordinates with back-up generation units to guarantee redundancy and resilience. The UPS serves as a buffer against power interruptions and grid disturbances, ensuring continuous and stable power delivery to servers and network infrastructures [21]. Within the data center, the infrastructure can be broadly categorized into two domains: 1) facility infrastructure, which includes cooling systems and auxiliary support equipment and 2) server and network infrastructure, which comprises of conventional servers (e.g., CPUs), accelerated servers (e.g., GPU-based systems), and other IT components.

A. DATA CENTER ELECTRICAL ARCHITECTURES

A data center's electrical architecture defines the outcome of its efficiency, reliability, and overall cost [22]. Fig. 5 illustrates the advancement of data center electrical architectures from the 208-V AC distribution system toward advanced 800-V DC configurations developed to support the increasing computational and power density requirements of AI data centers. The evolution highlights a gradual transition from legacy AC-based topologies with multiple power conversion stages to modern hybrid AC/DC and fully DC-distributed architectures, aimed at reducing conversion losses, improving efficiency, enhancing scalability, and supporting high-performance

TABLE 3. Comparison of AC and DC Power Architectures in Data Centers

Architecture	Advantages	Disadvantages
AC Power	Mature standards; wide compatibility with utility grids; well-established protection and control schemes	Multiple power conversion stages increase losses; lower overall efficiency; higher system complexity
DC Power	Higher efficiency due to fewer conversion stages; easier integration with renewable energy sources and energy storage; improved power quality and controllability	Limited standardization; higher initial cost; limited fault tolerance and reduced robustness under fault conditions; protection and fault isolation challenges; compatibility issues with legacy AC equipment.

computing infrastructures. The architecture includes the full arrangement of AC power delivery elements and how they are connected from the utility to the UPSs, backup generators, ESSs, PDUs, and down to the power supply unit (PSU). In this setup, the UPS serves as a protection scheme for sensitive loads, ensuring continuous operation by conditioning the incoming supply and providing short-term backup during outages [23]. The PDUs then distribute and monitor this conditioned power throughout the facility, stepping it down or transforming it as needed to match the requirements of different equipment zones [24]. At the final stage, the PSUs within each server convert the incoming AC power to an LV DC level required by the electronic components, while also managing power quality and supporting energy-efficient operation.

Over time, the rise in energy costs and the push toward sustainability led data center electrical designers to investigate alternative power distribution architectures [25]. Among these, DC distribution gained significant attention due to its successful deployment in data centers over the past decade, offering notable advantages such as improved energy efficiency, reduced conversion losses, enhanced power quality, and increased system reliability [26]. Despite these benefits, the widespread adoption of DC architectures remained limited because of the lack of universally accepted standards, as well as the challenges associated with specialized protection requirements and complex fault-coordination strategies in DC-based systems [22].

In response, more advanced and efficient power conversion technologies were developed to minimize losses, forming the basis of what is now considered the traditional AC architecture widely deployed in traditional data centers. This approach has remained adequate for legacy and traditional facilities with moderate rack-level power densities. However, the rapid growth of AI workloads has led to an unprecedented increase in computer-rack power density, prompting designers to reevaluate the suitability of existing architectures and to reconsider DC-based systems. Recently, NVIDIA [27] has released a technical report proposing an 800-V DC architecture tailored for next-generation AI infrastructure, illustrating how high-voltage DC distribution can be integrated with ESSs to meet the demands of AI data centers. Table 3 provides a comparison of AC and DC power architectures in data centers, which highlights their respective advantages, disadvantages, and suitability for large-scale AI data centers. Furthermore, Fig. 4 demonstrates that increasing the operating distribution voltage significantly enhances the power transfer capability

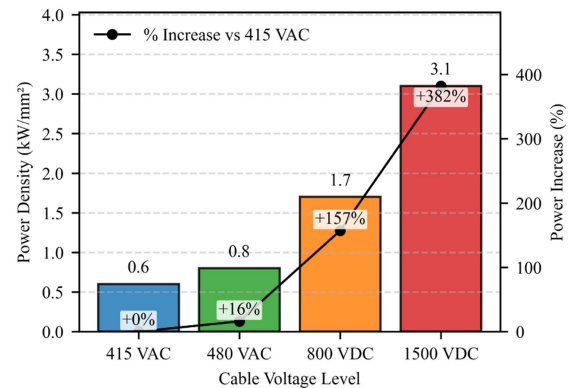


FIGURE 4. AC versus DC power cable comparison; data extracted from [27].

and cable power density of data center power systems. Compared with the conventional 415-V AC architecture, higher voltage AC and especially DC distribution systems can deliver substantially more power through the same conductor size, with 800- and 1500-V DC architectures providing major improvements in transferable power and cable utilization. This shows that higher voltage DC architectures can better support the growing rack-level power demands of large-scale AI data centers. Thus, this subsection reviews and compares conventional AC and emerging DC-based power distribution approaches, which differ significantly in terms of power distribution topology, conversion stages, and overall infrastructure demands.

1) CONVENTIONAL CPU DATA CENTERS

Data center electrical architectures mainly use AC distribution systems. These systems were designed to support general IT, enterprise computing, and cloud workloads, with rack-level power densities typically ranging from 3 to 10 kW [28].

Fig. 5(a) and (b) illustrates the AC distribution electrical architecture that has been widely adopted in legacy and traditional data centers, respectively. As shown in Fig. 5(a), utility power enters through the MV bus and is stepped down to the LV distribution level, where it supplies both the IT load and the facility infrastructure, including chillers, CRAH units, pumps, and lighting. A double-conversion LVAC UPS forms the core of the critical power path, providing voltage conditioning and ride-through capability through an AC–DC–AC conversion chain and an integrated battery system. At the downstream side of the UPS, power is further transformed

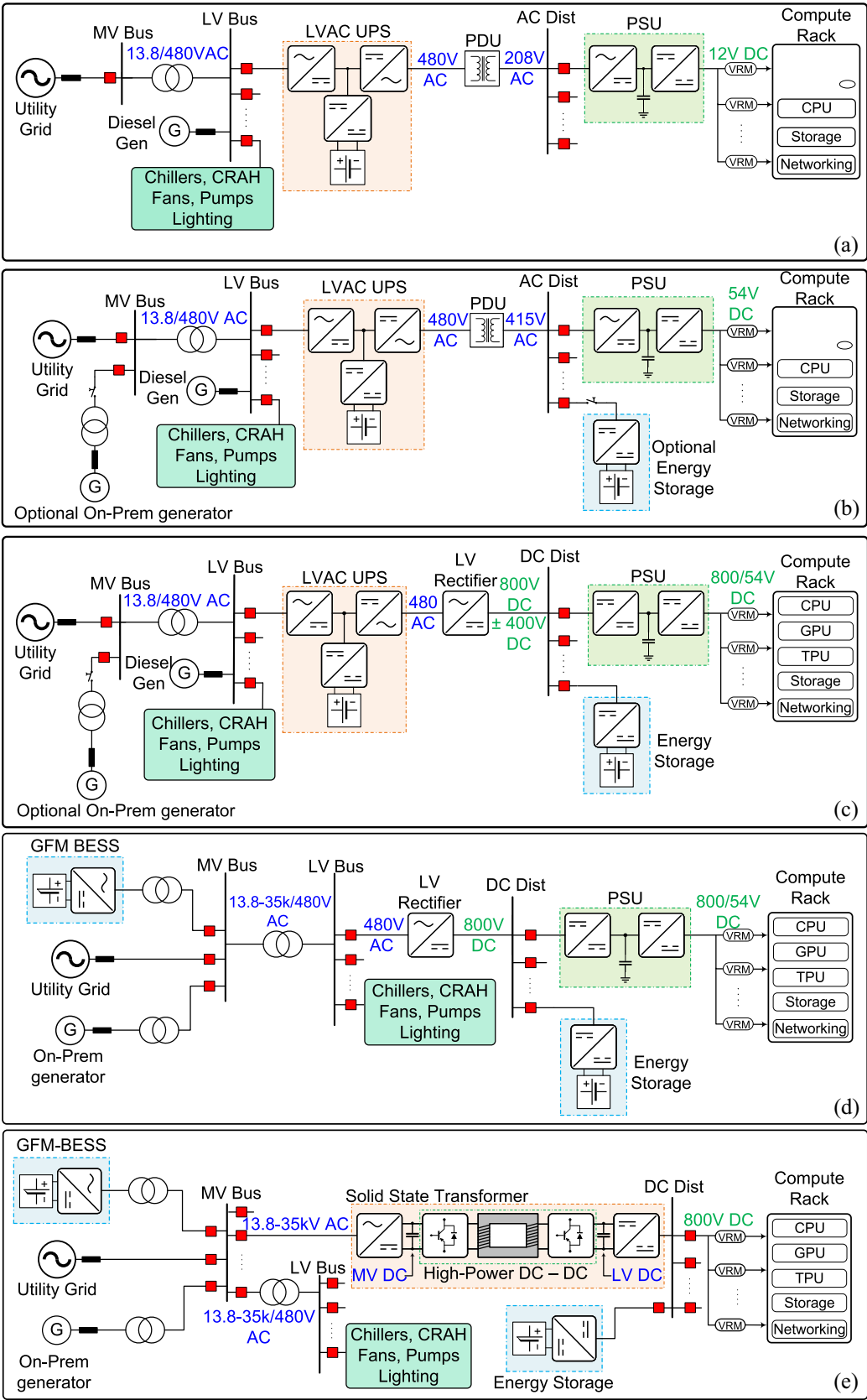


FIGURE 5. Advancement of data center electrical architectures. (a) Legacy, (b) traditional, (c) modern data center distribution with white-space retrofit, (d) modern data center hybrid power distribution, and (e) future-ready power distribution architecture [27], [33].

by rack-level or floor-level PDUs (e.g., 415/208 V AC), after which server PSUs perform an additional AC–DC conversion to deliver 12 V DC to individual VRMs that feed the CPU, storage, and networking modules.

The proliferation of virtualization and hyperscale computing brought about the adoption of modular and redundant architecture. Fig. 5(b) illustrates the existing traditional architecture, which represents an improvement over legacy systems. This architecture incorporates optional generators for redundancy and supports limited grid-interactive operations. In addition, external energy storage units are integrated to help smooth the fast power dynamics caused by GPU workloads [27]. It is worth noting that the voltage supplied to the rack is 415 V AC, while the PSU voltage is increased to 54 V DC. This incremental improvement reduces conversion losses and increases the rack power density. One of the key advantages of this architecture is the improvement in PUE values, which have decreased.

Conventional AC-based architectures remain robust, mature, and widely compatible with grid infrastructure. Nonetheless, their multiple conversion stages inherently limit efficiency and scalability, making them less suitable for modern high-density AI workloads.

2) ACCELERATED GPU DATA CENTERS

The emergence of AI technologies and GPU-accelerated computing has fundamentally altered the power delivery requirements of modern data centers. Contemporary AI training clusters exhibit rack-level power densities exceeding 10–30 kW [28], with next-generation GPU superpods trending toward 80–120 kW per rack [29], which is far beyond the capacity design for conventional AC architectures. These high-density systems exhibit rapid and unpredictable load transients during training cycles, placing significant stress on distribution equipment, power conversion stages, and thermal infrastructure [30]. As a result of these challenges, new transitional architectures are being designed to bridge the gap between the conventional AC power distribution architecture and the long-established ± 400 V DC architecture widely used in telecommunication and data centers [31], as well as the emerging 800-V DC data center architecture shown in Fig. 5(c) [27]. Although ± 400 V bipolar DC architectures theoretically provide inherent advantages in grounding and fault isolation [27], their deployment is constrained by the lack of protection devices. However, to relieve the growing imbalance between infrastructure and compute space caused by AC power's multiple conversion stages [32], transitional 800-V DC side-power racks housing localized rectifiers and distribution equipment can be introduced to shift conversion out of the compute rack and enable higher density AI deployments. Therefore, in this architecture, the facility-level AC–DC conversion is centralized, and the $\pm 400/800$ V DC bus replaces the AC distribution stage, with racks receiving high-voltage DC directly and performing only DC–DC conversion. This approach does

not provide a fully optimized end-to-end solution, as it introduces additional conversion stages [27]. Nevertheless, it offers a practical upgrade pathway from traditional AC infrastructure toward a high-voltage DC architecture without requiring immediate large-scale electrical room replacements.

A related approach, as illustrated in Fig. 5(d), involves a facility-level hybrid AC–DC power architecture designed to support high-density computing environments, including racks for AI accelerators. The architecture starts at the MV interface, where utility power, optionally supplemented by an on-site generator, is stepped down through MV/LV transformers to create a 480-V AC LV bus. A key feature of this architecture is the implementation of a facility-level AC/DC rectification stage. Multiple rectifier strings can operate in parallel, allowing for modular expansion and providing a stable DC bus that can support multimegawatt IT loads (approximately 1.5 MVA) [27]. The architecture enables seamless integration of energy storage units on the DC side. These storage systems, connected directly to the DC bus, offer rapid buffering for voltage support, transient suppression, and ride-through capability. Importantly, their direct DC connection avoids additional conversion losses and ensures a subsecond response to the fast load fluctuations typical of GPU workloads. Overall, this architecture provides a flexible foundation for integrating fast-acting energy storage particularly suited for emerging AI data center deployments, where high power densities, rapid transients, and stringent efficiency requirements exceed the capabilities of traditional AC-only architectures.

To support the long-term growth and rising power demand of AI data centers, Huntington and Tu [27] evaluate the use of MV rectifiers for converting MV AC input to an 800-V DC power feed. The study also examines solid state transformer technology as a potential future option for enhancing resilience and distribution efficiency in next-generation facilities. As shown in Fig. 5(e), the power conversion section takes MV AC (up to 35 kV) and directly converts it into 800 V DC using a solid state transformer conversion chain. This chain integrates AC-to-DC rectification, large-scale DC–DC voltage regulation, and high-capacity DC distribution specifically designed for dense AI compute racks. By eliminating the intermediate 480-V AC distribution layer, this design shortens the power path and simplifies infrastructure, reducing complexity.

B. COOLING SYSTEM ARCHITECTURES

Effective thermal management is fundamental to ensuring that equipment in modern data centers operates reliably and efficiently. As rack power densities continue to increase and heat loads intensify, the efficiency of heat rejection and thermal control mechanisms has become a defining factor in the overall performance [34]. The effectiveness of a cooling system, whether implemented at a rack, row, or facility scale, is influenced by multiple factors, including airflow distribution, heat transfer efficiency, coolant type and flow characteristics, system integration with IT load dynamics, and

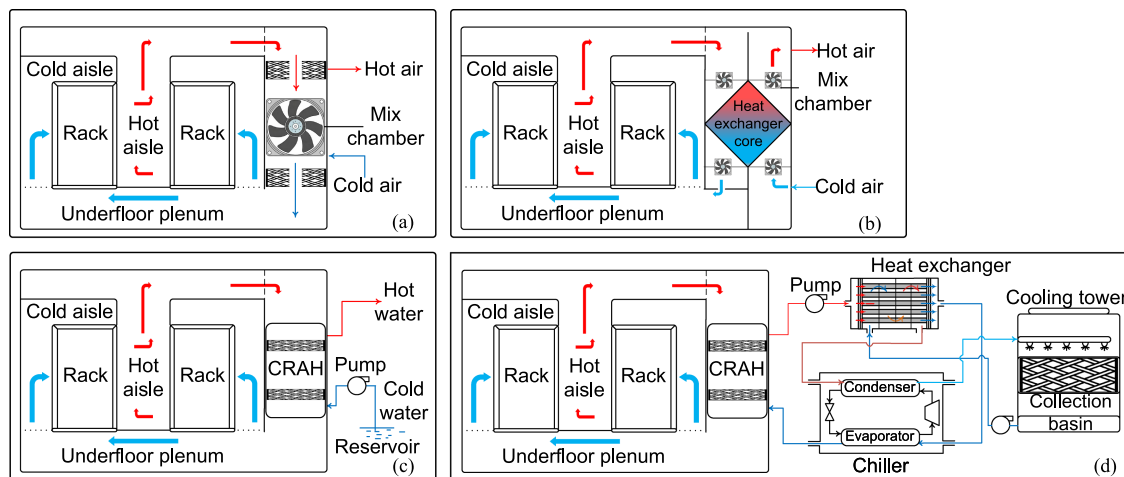


FIGURE 6. Schematic diagram of direct and indirect free cooling architectures. (a) Direct air-side cooling. (b) Indirect air-side cooling. (c) Direct water-side cooling. (d) Indirect water-side cooling [35].

the resilience of components under varying environmental conditions [35]. To advance system efficiency, a wide range of cooling architectures have been developed, focusing primarily on advancements in cooling system, thermal interface design, and control strategies. These innovations directly influence heat dissipation efficiency, PUE, and long-term operational reliability.

Among the various cooling configurations deployed in data centers, system architectures can be broadly categorized into air-based, liquid-based, and hybrid cooling frameworks [35], [36]. As implied by the term, air-based cooling relies on conditioned airflow to dissipate heat, though numerous design optimizations such as hot/cold aisle containment, in-row cooling, and raised-floor plenum systems have been explored to enhance thermal performance. In contrast, liquid-based cooling solutions are distinguished by their direct or indirect contact mechanisms, ranging from liquid immersion cooling to cold-plate and rear-door heat exchanger systems, while hybrid cooling approaches combine both air and liquid cooling techniques to improve thermal efficiency, operational flexibility, and energy effectiveness. The following subsection provides a detailed analysis of these cooling architectures, examining their structural configurations, operational advantages, and limitations in modern and next-generation data center environments.

1) AIR COOLING SYSTEMS

Air-based cooling remains the most widely adopted thermal management approach in conventional data centers due to its mechanical simplicity, ease of deployment, and compatibility with existing IT hardware layouts. In typical configurations, heat generated by IT equipment is transferred to the surrounding air and transported out of the white space through controlled airflow pathways. The performance of such systems is determined by the efficiency of air distribution, pressure management, and the ability to minimize thermal mixing between hot and cold air streams. Based on

architectural level, air cooling is commonly implemented at the room, row, and rack levels, with the choice largely determined by the heat density of the IT equipment. Room-level systems rely on CRAC or CRAH units that deliver cold air through raised-floor plenums to cold aisles, enabling a “face-to-face, back-to-back” rack configuration that separates cold and hot airflow streams. While this architecture is simple and widely deployed, its long airflow paths and large thermal management domain make it susceptible to mixing losses and nonuniform cooling [35]. To address these limitations, row-level cooling positions dedicated CRAH units directly alongside specific rack rows to shortening the airflow distance and improving the targeting of cold air. At the most targeted level, rack-level cooling integrates the cooling unit directly within the rack, enabling heated air to return directly to the cooling module rather than recirculating into the room. This configuration supports much higher heat densities, typically above 50 kW per rack, while delivering more precise airflow control and greater cooling capacity utilization.

Free cooling is an advanced and well-established outdoor cooling strategy used in conventional and large-scale data centers and is particularly effective in reducing mechanical cooling energy demand [17]. The effectiveness of this cooling architecture has been validated across a wide range of climatic and operational studies. For instance, analyses have shown that the technique performs optimally in regions with ambient temperatures below 20°C and relative humidity above 60% [37]. Thus, seasonal benefits of optimal control frameworks can reduce the total energy of the facility energy during winter. Performance potential also exhibits strong geographical variability; some temperate and coastal cities are capable of operating in free cooling mode for more than 3000 hours annually, whereas colder regions increasingly rely on flexible decision thresholds to accommodate wider environmental operating ranges [38].

This approach is commonly implemented through air-side and water-side cooling strategies, which can be classified

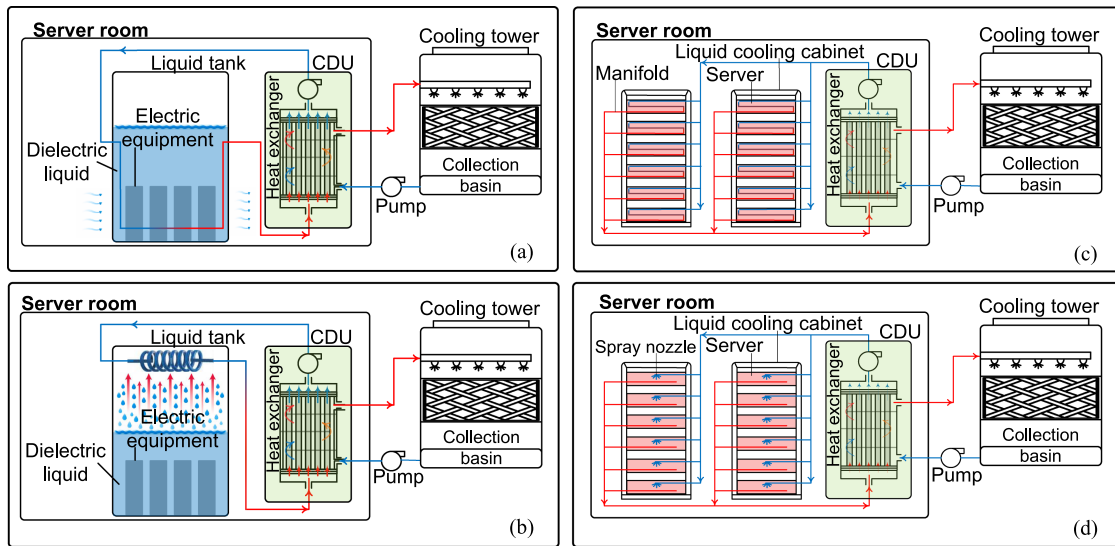


FIGURE 7. Schematic diagram of liquid cooling architectures. (a) Single-phase immersion liquid cooling. (b) Two-phase immersion liquid cooling. (c) Cold plate liquid cooling. (d) Spray liquid cooling [39].

as direct or indirect free cooling. Direct air-side cooling, shown in Fig. 6(a) introduces filtered outside air directly into the server room to remove heat, offering high efficiency but relying heavily on ambient weather conditions. Indirect air-side cooling, shown in Fig. 6(b), keeps outdoor air separate from indoor air by using heat exchangers, improving control and minimizing contamination risks. For water-side strategy, direct cooling (water-side) circulates chilled water or coolant directly through cooling coils inside the server room to absorb heat from the recirculating air, as shown in Fig. 6(c). Indirect water-side cooling, shown in Fig. 6(d), transfers heat through intermediate loops such as cooling towers, chillers, or heat exchangers without bringing facility water into direct contact with IT equipment environments, allowing safer operation and easier integration with free cooling methods.

Overall, free cooling remains a climate-dependent optimization method, with real-world energy benefits tightly coupled to localized atmospheric conditions and site-specific inlet-air constraints. However, conventional air cooling approaches are physically insufficient to handle the extreme heat densities of AI data centers.

2) LIQUID-BASED COOLING SYSTEMS

Modern processors can exceed 100 W/cm^2 , while air cooling typically dissipates only around 37 W/cm^2 , creating a widening thermal gap that leads to inefficiency, rising energy use, and thermal instability in data centers [17]. This challenge is intensified by the rise in rack power density, with next-generation deployments expected to exceed 30 kW per rack, compared with much lower densities in traditional systems. To tackle these escalating demands, liquid cooling has emerged as a leading solution and a key direction for next-generation data center infrastructure.

Recent research consistently highlights the performance and efficiency advantages of liquid cooling in AI data centers. Ramakrishnan et al. [40] report that direct liquid cooling can improve GPU performance efficiency by 2.7%, cut power consumption by 12%, and reduce chip temperatures by roughly 20°C . These results are reinforced by Latif et al. [41], who show that liquid-cooled GPUs operate at significantly lower temperatures (41°C – 50°C) than air-cooled counterparts (54°C – 72°C), enabling up to 17% higher performance. As processor thermal design power trends toward 700 W , Azarifar et al. [42] argue that liquid cooling is becoming essential for sustaining performance and energy efficiency in modern AI-driven data centers. Major energy companies such as Shell, ExxonMobil, and BP are increasingly supporting the adoption of liquid cooling in data centers, backed by several pilot initiatives and strong projected market growth [43]. Liquid cooling technologies for servers generally fall into three categories: immersion cooling (single- and two-phase), cold-plate cooling, and spray-based cooling [39]. Among these, cold-plate cooling has become the most widely adopted approach. As noted by Zhang et al. [44], it is the earliest established and most commonly used liquid cooling method, and it delivers significantly better thermal performance than conventional air cooling.

Fig. 7(a) shows a single phase immersion cooling, where servers are fully submerged in a liquid coolant that always remains in the liquid state, absorbing heat through sensible heat transfer before being pumped to a heat exchanger. This approach uses a simple control design, limits coolant loss through effective sealing, and requires little to no fluid replacement. In contrast, as shown in Fig. 7(b), two-phase immersion cooling submerges servers in a low-boiling-point dielectric fluid that vaporizes when the server components heat up, with the vapor then condensed and returned to the tank. By using latent heat during this phase change, this

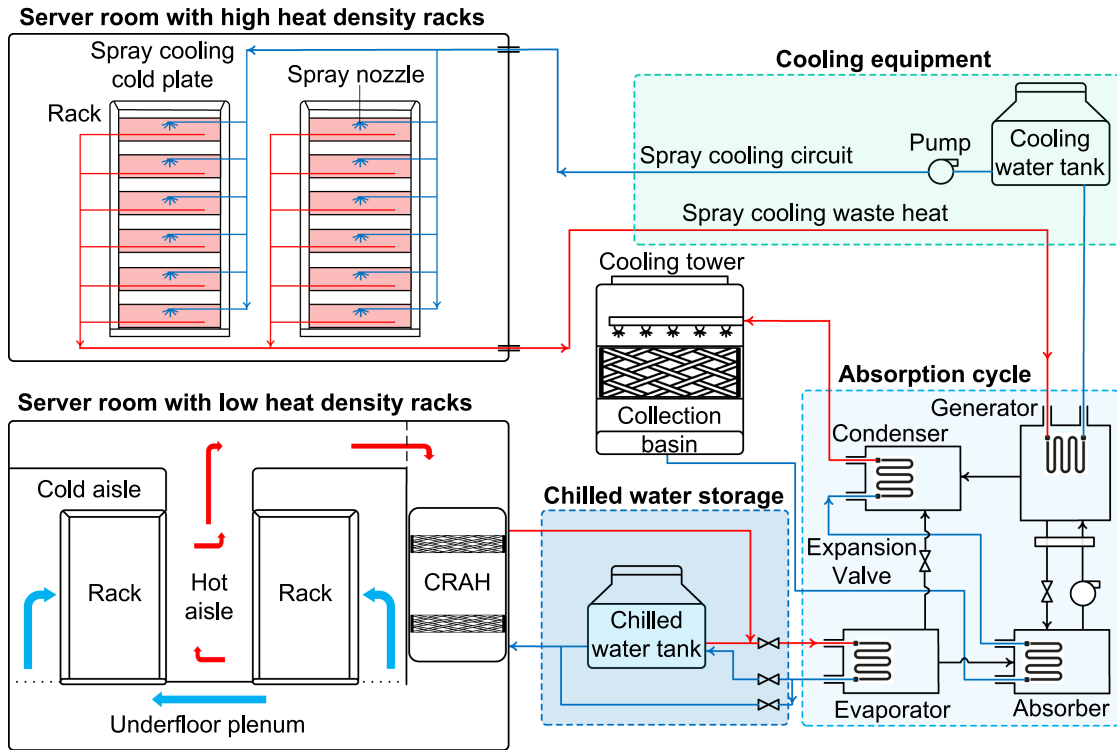


FIGURE 8. System configuration of a hybrid cooling architecture for future AI-driven data centers [36].

approach can be used to reject heat efficiently. In cold-plate liquid cooling, the cooling targets high-heat components like CPUs and GPUs by transferring heat to a circulating coolant through attached cold plates, as shown in Fig. 7(c). The coolant flows through a manifold to each server and then to a CDU, which rejects heat to an external cooling system. Finally, spray liquid cooling delivers coolant directly onto heat-producing components through precision nozzles, using less fluid than immersion systems, as shown in Fig. 7(d). The warmed coolant is collected, cooled, and recirculated through a system that includes a pump, heat exchanger, and external cooling unit.

3) HYBRID COOLING SYSTEMS

The hybrid cooling architecture integrates both air-based and liquid-based cooling to support the high heat densities typical of AI data centers [1]. As illustrated in Fig. 8, the approach proposed by Chen et al. [36] can be adopted, where spray-cooled cold plates are used for high-density compute racks. In this setup, liquid coolant removes heat directly at the chip level and transfers it to a cooling water tank and subsequently to the cooling tower. Air conditioning systems can be used for lower heat density racks, which circulate conditioned air through the servers. A chilled-water storage system, powered by an absorption cycle comprising of an absorber, generator, evaporator, and condenser, further enhances cooling flexibility and stabilizes thermal performance during fluctuating workloads. Coordinated operation of these subsystems can enable efficient heat rejection and maintains reliable temperature control under varying computational conditions. This can be

used to avoid the significant cost and complexity associated with fully liquid-cooled AI data center designs.

IV. MODELING OF GRID-INTEGRATED DATA CENTERS

Modeling is essential for understanding the dynamic interactions between grid-integrated data centers and modern power systems. Different modeling approaches are required depending on the intended study objective, ranging from steady-state power flow analysis to fast EMT and stability investigations. In addition, converter-driven phenomena such as harmonics, resonance, and interaction with weak-grid introduce further challenges that necessitate detailed representation of converter dynamics and network behavior. This section discusses the major modeling approaches for grid-integrated data centers and examines the key converter-driven phenomena that influence system performance and stability.

A. MODELING APPROACHES

Modeling approaches are fundamental to the analysis of grid-integrated data centers, as they provide mathematical and computational representations of electrical components, control systems, and their interactions to analyze, predict, and optimize system behavior under a wide range of operating and disturbance conditions [45]. In this context, modeling enables the systematic study of the tightly coupled dynamics between data center server workloads, power supply distribution infrastructures, cooling systems, energy storage, on-site generation, and the external power grid. By capturing essential electrical and operational characteristics while simplifying

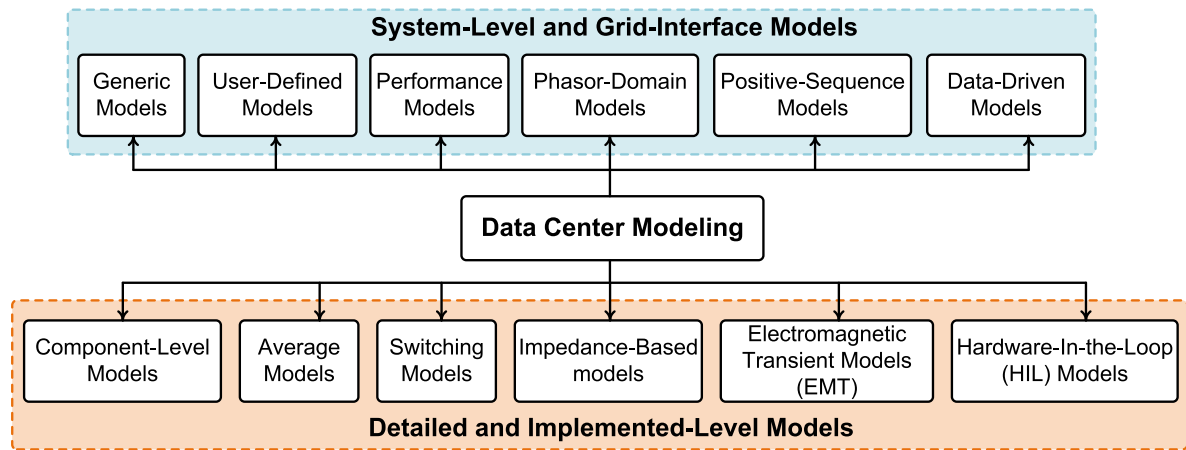


FIGURE 9. Modeling approaches for data centers.

noncritical details, these models can support the investigation of key phenomena such as power flow, stability, control, reliability, and flexibility across different time scales [46]. Therefore, the level of model fidelity should be selected based on the intended application, striking a balance between accuracy and computational efficiency [47], [48].

This subsection reviews the main modeling approaches shown in Fig. 9 for grid-integrated data centers, ranging from detailed device-level models for component and transient analysis to simplified system-level and grid-interface models used for stability and large-scale power system studies.

1) GENERIC MODELS

Generic models are widely utilized for power system modeling, particularly as reduced-order models in transmission planning that involves dynamic components. These models are better suited for long-term planning and high-level flexibility assessments than for detailed stability or protection analysis. This approach can also effectively represent grid-integrated data centers by focusing on aggregated power demand, energy flows, and flexibility characteristics, while often neglecting model fidelity and specific manufacturer details [49]. For example, Ounifi et al. [50] introduce a generic data center metamodel capable of representing heterogeneous infrastructure characteristics. Such an abstract model approach allows for scalable analysis of system-wide impacts, including grid planning, capacity adequacy, and demand response potential under varying operating conditions. While generic models provide computational efficiency and conceptual clarity, their reliance on lumped parameters limits their ability to capture detailed dynamic interactions between internal subsystems, especially during rapid disturbances or control-driven transients [51].

2) USER-DEFINED MODELS

User-defined models provide researchers with the flexibility to tailor system representations according to specific architectural choices, control strategies, or research objectives.

Research highlights that, unlike standardized modeling frameworks, these models explicitly enable granular exploration of server behavior, power electronics, cooling strategies, or energy management algorithms, allowing investigation of novel data center grid interaction paradigms [52]. Another research proposed a modular toolbox that allows building data center models of any scale and configuration using standardized building blocks [53]. These models offer detailed representations and are more precise than generic ones. However, they are more complex to tune, resulting in slower simulation performance. Consequently, while user-defined models are powerful tools for innovation-driven research, their applicability to broader system studies requires careful documentation and validation against real-world data.

3) PERFORMANCE MODELS

Performance models aim to establish quantitative relationships between electrical operating conditions and data center service-level metrics, such as throughput, latency, or computational efficiency under different load characteristics [54]. Within this context, Kühn [55] developed energy efficiency performance models that relate server operations to SLAs, while Schlitt and Nebel [56] proposed the load-dependent energy efficiency metric, integrating utilization, performance, and power models to provide detailed efficiency characterization. These studies demonstrate that such models can facilitate the integration of workload characterization with power consumption, enabling the evaluation of how grid events or power constraints affect computational performance. For grid integration studies, performance models are useful for evaluating demand response strategies by balancing temporary power modulation with quality-of-service requirements [57]. Their fast execution and low data requirements make them suitable for early-stage or high-level analyses. However, their reliance on empirical relationships limits accuracy under unforeseen conditions, and they lack the fidelity needed to capture electrical transients or converter-level interactions.

4) COMPONENT-LEVEL MODELS

Component-level models represent individual physical elements within the data center, including servers, PDUs, uninterruptible power supplies, converters, cooling equipment, and ESSs [54]. By modeling the electrical and thermal characteristics of each component, these approaches enable detailed investigation of internal power flows and loss mechanisms [58]. This modeling approach is utilized in [59] and [60] for reliability studies, failure mode analysis, and detailed control design, where it is essential to understand the interactions between specific components. Although component-level modeling provides high fidelity and physical insight, its application to large-scale systems can be computationally intensive, often necessitating model reduction or aggregation techniques when studying interactions with the grid.

5) PHASOR-DOMAIN MODELS

Phasor-domain models represent voltages and currents using fundamental-frequency phasors, thereby capturing steady-state and slow dynamic behavior while neglecting high-frequency switching phenomena. These models are a cornerstone of conventional power system analysis and have been increasingly applied to study the grid impact of large data centers, particularly in terms of voltage regulation, power flow, and frequency stability [61]. Several studies, including [59] and [62], adopt phasor-domain models for grid-integrated data center studies to support efficient large-scale network simulations, making them suitable for contingency assessment and operational planning. However, the inherent assumption of sinusoidal steady-state operation limits their ability to capture fast control interactions, harmonics, and EMTs associated with power electronic interfaces [63] prevalent in AI data centers.

6) POSITIVE-SEQUENCE MODELS

Positive-sequence models represent three-phase power systems using only the balanced fundamental-frequency component, assuming symmetrical operating conditions and neglecting unbalanced and harmonic effects. This modeling approach is used for longer time scales, particularly for frequency response, and system-level stability and load-flow studies [64]. By significantly reducing model complexity, positive-sequence representations enable efficient simulation of extensive networks with high data center penetration, making them suitable for long-duration and wide-area studies, as used in [54] for studying UPS performance in a data center. However, considering the underlying assumptions of balanced operation limit their applicability in scenarios involving asymmetrical faults, unbalanced loading, or power-electronic-induced harmonics. Consequently, positive-sequence models are generally unsuitable for detailed power quality, protection, or electromagnetic interaction studies involving data center infrastructure [65].

7) EMT MODELS

EMT models provide a detailed time-domain representation of electrical systems, capturing fast dynamics, switching actions, and nonlinear control behavior [66]. Therefore, in the context of grid-integrated data centers, EMT models are essential for analyzing converter-dominated interfaces, protection coordination, and system response to faults or abrupt disturbances [67]. These models enable accurate assessment of high-frequency interactions between data center power electronics and the grid, which are increasingly important as data centers adopt advanced inverter-based architectures. Despite their high fidelity, EMT simulations are computationally demanding and typically limited to small subsystems or short simulation windows, restricting their use in large-scale or long-term studies [68].

8) HIL MODELS

HIL modeling integrates physical hardware with real-time simulation platforms to assess system behavior in controlled and repeatable test environments. By operating the hardware and simulator in a closed-loop configuration, this approach enhances the realism of system evaluation and enables detailed analysis of complex interactions that are difficult to capture using purely software-based simulations [69]. Sun et al. [70] employ a HIL approach by integrating a physical converter-based emulator with a real-time simulated model of the data center power distribution system. Given the growing need for dynamic data center models, this approach is essential for validating controller implementations, inverter hardware, and protection schemes under realistic grid disturbances. HIL modeling bridges the gap between simulation studies and field deployment [71]. However, their reliance on specialized equipment, real-time constraints, and limited scalability confines their use primarily to validation and prototyping rather than large-system analysis.

9) SWITCHING MODELS

Switching models represent the behavior of power electronic switching devices, together with their associated pulsewidth modulation (PWM) control schemes and gating circuits. By modeling the generation of sinusoidal voltage and current waveforms through high-frequency switching and embedded control structures, these models capture detailed converter dynamics, harmonic distortion, and complex control interactions with the surrounding power system [72]. With the challenges, such as frequency coupling and low-frequency oscillations, posed by AI data centers, switching models are essential for investigating issues such as harmonic distortion, converter-induced instabilities, and high-frequency interactions with the grid [73]. These models provide the highest level of electrical fidelity and are often used as reference benchmarks for validating reduced-order or average models. However, their extreme computational complexity restricts their application to small-scale systems or short-duration simulations, making

them impractical for large networks or long-term operational studies [72].

10) AVERAGE MODELS

Average models simplify the representation of power electronic converters by averaging switching behavior over a switching period, thereby retaining essential dynamic characteristics while eliminating high-frequency components. These models are commonly utilized in dynamic stability and control studies of grid-connected inverter-based resources, where switching-level detail is less [72], which now includes AI data centers. Average modeling significantly reduces computational burden compared to EMT or switching models, enabling longer simulation times and larger system representations. Nevertheless, the exclusion of switching harmonics and fast transients limits their applicability in studies involving electromagnetic compatibility, protection, or harmonic interaction analysis [74].

11) IMPEDANCE-BASED MODELS

The impedance-based modeling approach is gaining popularity for inverter-based electrical networks, such as inverter-based resources, dominant electricity grids, and microgrids. It is a frequency-domain-based technique used to analyze resonance stability issues in electrical networks. In impedance-based modeling, loads are represented by input impedance, while sources are represented by output impedance, and the ratio between the source and load impedance must satisfy the Nyquist stability criterion. This technique has been employed to analyze the resonance incident in a data center. The study has developed an impedance-based model for the PSU of a server and applied the Nyquist stability criterion for stability analysis [75], [76]. Thus, this modeling approach can be utilized to analyze the resonance-based stability issues in data centers, as recently employed by Ruan et al. [77] for data center control against subsynchronous resonance.

12) DATA-DRIVEN MODELS

Data-driven models leverage measurement data and machine learning techniques to capture the dynamic behavior of data centers without requiring detailed physical representations of internal components. By learning input–output relationships directly from data, a data-driven approach is used in [78] to simulate a data center cooling system and predict its responses hours into the future. Such models are beneficial for real-time control, forecasting, and system-level studies where detailed equipment parameters are unavailable. Despite their flexibility and lower computational demand [72], data-driven models often suffer from limited interpretability, reduced generalization outside the operating conditions represented in the training data, and data scarcity due to confidentiality constraints. Moreover, their performance is highly dependent on data quality [72].

The reviewed modeling approaches demonstrate that no single modeling approach is suitable for all applications, as

the choice solely depends on the required fidelity, computational complexity, simulation time scale, and the intended grid integration study. High-fidelity models provide detailed dynamic and transient behavior at the component level, while system-level models enable efficient large-scale power system analysis and control evaluation. A comparative summary of these modeling approaches, including their fidelity level, simulation time-step, advantages, and limitations, is presented in Table 4. Furthermore, Fig. 10 presents a model-selection workflow that links study objectives, simulation timescales, required parameters, and validation requirements to the most appropriate modeling approach, thereby providing a practical guideline for selecting suitable models for planning, operational, stability, and EMT studies of large-scale data centers.

B. CONVERTER-DRIVEN PHENOMENA

The interaction between converter-based loads and the power grid introduces a range of dynamic phenomena that can significantly influence power quality, system stability, and electromagnetic behavior. Therefore, accurate modeling of converter-driven interactions is essential for analyzing system stability, assessing grid impacts, and developing mitigation strategies under both normal and weak-grid operating conditions.

1) HARMONIC DISTORTION

The nonsinusoidal currents drawn by data center power electronics constitute one of the most dominant forms of converter-grid interaction. Modern server PSUs typically employ single-phase boost or dual-boost PFC converters at the front-end stage, whose switching-cycle averaged inductor current dynamics can be expressed as follows:

$$\frac{di_L}{dt} = \frac{v_a - (1 - d')v_{DC}}{L} \quad (1)$$

where v_a is the AC input voltage, d' is the off-time duty ratio, L represents inductance, and v_{DC} is the DC-link voltage. The bilinear term $(1 - d')v_{DC}$ introduces nonlinear behavior into the converter dynamics [79]. For this reason, even under ideal sinusoidal supply conditions, switching actions and imperfect current tracking generate harmonic components at odd multiples of the fundamental frequency. In large-scale data centers containing tens of thousands of PSUs operating simultaneously, these harmonic currents aggregate throughout the internal distribution hierarchy and propagate back toward the PCC. The resulting harmonic voltage distortion at the PCC can be approximated as

$$V_h = I_{DC,h} Z_h(\omega_h) \quad (2)$$

where $I_{DC,h}$ represents the aggregated harmonic current at harmonic frequency ω_h , and $Z_h(\omega_h)$ denotes the grid impedance at that frequency [19]. Although reactive power panels and line inductors provide partial attenuation of high-frequency distortion, these passive elements also contribute additional inductive impedance capable of interacting with converter

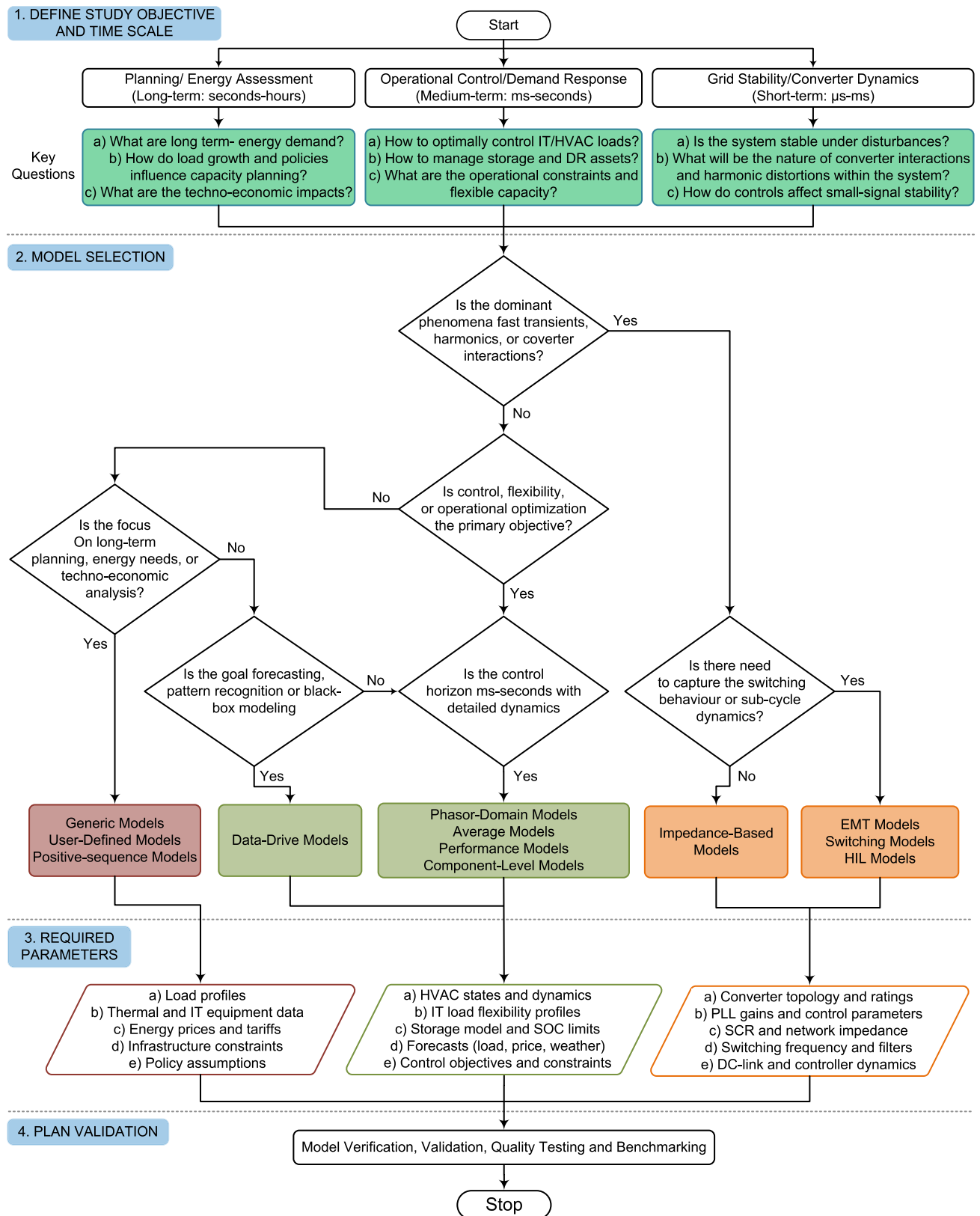


FIGURE 10. Flowchart for selecting data center models based on study objectives, timescales, and parameters.

TABLE 4. Summary of Advantages and Limitations of Data Center Modeling Approaches

Modeling Approach	Fidelity Level	Time Step	Advantages	Limitations
Generic Models	Low	ms–s	Very low computational cost; suitable for large-scale, long-term planning studies; easy integration into bulk power system simulations	Oversimplified representation of internal data center dynamics; cannot capture transient, control, or component-level behavior; limited accuracy under fast grid disturbances
User-Defined Models	Low–Medium	ms–s	High flexibility for novel architectures and control strategies; useful for exploratory and emerging research concepts	Lack of standardization complicates validation and comparison; strong dependence on modeling assumptions; reduced reproducibility across studies
Performance Models	Low–Medium	ms–s	Directly link electrical operation with IT performance and QoS; well suited for demand response and workload-aware control studies	Often based on empirical relationships; limited electrical and dynamic fidelity; unsuitable for transient, protection, or stability analysis
Component-Level Models	High	μ s–ms	High physical accuracy of devices and subsystems; enable detailed loss, reliability, and constraint analysis; support advanced control and failure studies	High modeling and computational complexity; poor scalability for large data center or grid-wide studies
Phasor-Domain Models	Medium	ms–s	Computationally efficient for large systems; suitable for power flow, voltage, and frequency stability analysis; widely supported in commercial simulation tools	Neglect high-frequency dynamics and switching effects; cannot capture EMT or harmonics
Positive-Sequence Models	Low	s	Highly efficient for large-scale transmission-level studies; effective for steady-state, frequency response, and contingency analysis; well supported by industry-grade tools	Assume balanced three-phase operation; cannot capture unbalance, harmonics, or switching dynamics; unsuitable for power quality or protection analysis
EMT Models	Very High	ns– μ s	Capture detailed switching, nonlinear, and fast control dynamics; essential for fault, protection, and converter interaction studies	Very high computational burden; limited scalability and simulation duration
HIL Models	Very High	real-time	Incorporate real hardware behavior and delays; enable realistic validation of control and protection schemes; reduce deployment and commissioning risk	Require specialized hardware and real-time simulators; limited scalability; primarily suited for testing rather than system-level analysis
Average Models	Medium	μ s–ms	Preserve essential converter dynamics with reduced complexity; suitable for stability and control studies over long horizons	Do not capture switching harmonics or fast transients; limited applicability for power quality and protection studies
Switching Models	Very High	ns–ms	Highest electrical fidelity; accurately represent harmonics and switching transients; useful as benchmarks for reduced-order models	Extremely computationally intensive; impractical for large systems or long-term simulations; model complexity reduces interpretability
Impedance-Based Models	Medium	frequency domain	Effective for resonance and oscillatory stability analysis; useful for converter–grid interaction studies	Requires accurate identification; complexity increases with size
Data-Driven Models	Variable	model-dependent	Capture nonlinear behavior without detailed physical modeling; suitable for forecasting, control, and operational flexibility studies; effective when internal data center details are unavailable	Limited physical interpretability; strong dependence on data quality and operating range; poor extrapolation under unseen or fault conditions

filter capacitance [76]. A more severe power quality issue emerges when the equivalent distributed capacitance of large converter populations resonates with the inductive grid impedance. Since server-level PFC filters exhibit a predominantly capacitive characteristic, the combined impedance between the distributed capacitance C_p and inductive components L_s becomes

$$Z_h = \frac{j\omega_h L_s}{1 - \omega_h^2 L_s C_p}. \quad (3)$$

As the denominator approaches zero, resonance conditions amplify even relatively small harmonic currents into significant voltage distortion [19]. This phenomenon may result in transformer overheating, increased system losses, insulation stress, and wideband oscillations spanning frequencies from several hertz to multiple kilohertz. Unlike conventional integer-order harmonics, these distortions are determined by

the interacting network inductance and distributed converter capacitance. AI data centers further intensify power quality challenges through rapid and stochastic load variations associated with AI intensive workloads. Fast transitions between compute and communication phases can produce subsecond power fluctuations that aggregate into large facility-level ramp events, causing voltage flicker, voltage sags, and transient harmonic amplification at the PCC [19].

2) RESONANCE

Resonance-induced oscillations arise from interactions among converter control systems, passive network elements, and grid impedance. These interactions can produce instability in both low- and high-frequency regions, leading to harmonic amplification, oscillatory behavior, and degraded system performance. Such phenomena are commonly analyzed using impedance-based stability methods, which

evaluate the dynamic relationship between source and load impedance across different frequency ranges. The medium-frequency PFC impedance model is

$$Z_{am}(s) = \frac{sL + V_{DC}H_c(s)}{1 + GV_{DC}H_c(s)} \quad (4)$$

where $H_c(s)$ denotes the current controller, G is the converter forward gain, and V_{DC} is the DC-bus voltage [75]. Low-frequency instability has been observed in several Meta data centers, where oscillations near 11 Hz were accompanied by strong sideband components at 49 and 71 Hz around the 60-Hz fundamental frequency [75]. In this frequency range, (4) is unable to accurately reproduce this phenomenon because it assumes a constant DC-link voltage and neglects DC-bus dynamics. In practice, perturbations at the AC terminal interact with the DC-link control loop through the bilinear coupling term introduced in (1), producing additional frequency-coupled current components [79]. To capture this behavior, the converter small-signal model must include three coupled transfer admittances

$$Y_a(s) = \frac{\hat{i}_a(s)}{\hat{v}_a(s)} \quad (5)$$

$$Y_{c-2}(s) = \frac{\hat{i}_a(s - j2\omega_1)}{\hat{v}_a(s)} \quad (6)$$

$$Y_{c+2}(s) = \frac{\hat{i}_a(s + j2\omega_1)}{\hat{v}_a(s)} \quad (7)$$

where $\hat{i}_a$ is the current response, $Y_a(s)$ denotes the input admittance of the PFC converter under ideal AC-source conditions, and $Y_{c-2}(s)$ and $Y_{c+2}(s)$ represent the associated transfer admittances [75]. At higher frequencies, the converter digital control delays become a dominant source of instability. For converters employing trailing-edge modulation, the effective control delay can be approximated as

$$T_d = 2f_1 \int_0^{\frac{1}{2f_1}} T_s \left[1 + \frac{d(\tau)}{2} + n \right] d\tau \quad (8)$$

where T_s is the switching period, $2f_1$ is the second harmonic frequency, d is the duty ratio of the switch, and n represents additional processing cycles [75]. Incorporating this delay into the converter impedance model (4) yields

$$Z_{ah}(s) = \frac{sL + V_{DC}e^{-sT_d}H_c(s)}{1 + GV_{DC}e^{-sT_d}H_c(s)}. \quad (9)$$

These interactions can produce high-frequency resonance and negative damping, particularly in systems with tightly regulated parallel converters [80]. In large data centers, instability mechanisms are not confined to the PCC because the internal facility impedance may substantially exceed the external utility impedance [76]. Consequently, oscillatory behavior can originate within the internal radial distribution network itself. To represent this behavior, the equivalent source impedance

framework aggregates impedance contributions across the hierarchical distribution structure

$$Z_s(s) = Z_0(s) + \sum_{i=1}^6 \left[Z_i(s) \prod_{j=1}^i N_j \right] \quad (10)$$

where N_j is the number of sublines powered from the line designated by impedance Z_j , $j = 1, 2, \dots, 6$. The impedance accumulation extends from rack-level distribution through reactive power panels, switchboards, transformers, and upstream substations. Stability assessment can then be performed using the Nyquist criterion applied to the loop gain formed by the converter admittance and the equivalent source impedance.

These equations are important for treating converter-driven resonance phenomena for large-scale data centers because their highly concentrated power electronic infrastructure causes complex interactions between converters, network impedance, and harmonic currents that can destabilize both the facility distribution system and the utility grid.

3) STABILITY IN WEAK GRID

Data centers exhibit largely passive impedance characteristics, dominated by tightly regulated switch-mode PSUs, UPS converters, and battery interfaces whose control loops can introduce negative damping and fast electromagnetic interactions [75], [76]. As a result, voltage stability at the PCC becomes highly sensitive to rapid active and reactive power fluctuations associated with AI workloads. The voltage variation at the PCC can be represented using voltage sensitivity relationships as

$$\Delta V_i = S_{VP,i} \Delta P_i + S_{VQ,i} \Delta Q_i \quad (11)$$

where $S_{VP,i}$ and $S_{VQ,i}$ represent the voltage sensitivity factors associated with active and reactive power variations [81]. For a radial connection, the PCC voltage deviation is approximately

$$|V_{PCC}| \approx |V_g| - \frac{R_{line}P_{L,DC} + X_{line}Q_{L,DC}}{|V_{PCC}|} \quad (12)$$

where the active and reactive power consumed by the data center are represented as $P_{L,DC}$ and $Q_{L,DC}$, respectively, and $|V_g|$ is the voltage magnitude of the source [19]. This indicates that under weak-grid conditions, where network impedance is high, significant voltage drops can occur during rapid load changes, leading to voltage instability. In addition, these load transients can excite frequency and rotor-angle dynamics across the wider power system. When large-scale AI workloads start, stop, or transfer to UPS operation during disturbances, the resulting power imbalance propagates through nearby synchronous generators according to the swing equation

$$\begin{aligned} \frac{d}{dt}(\Delta\omega) = & K (P_{ref} - P_{L,DC}(t)) \\ & + \left(\frac{3K|E_g||E|}{X} \right) \sin(\delta) - \alpha \Delta\omega \end{aligned} \quad (13)$$

where K is the generator inertia constant, α is the damping coefficient, $\Delta\omega$ is the frequency deviation, P_{ref} is the three-phase generator reference power, δ is the rotor angle, E_g is the internal generator voltage magnitude, E is the bus/grid voltage magnitude, and X represents the equivalent reactance between the generator and the grid. The term $(3|E_g||E|/X)$ represents the electrical power transfer interaction between the generator and the connected power system [19]. Accordingly, if the data center load changes faster than conventional generators can respond, frequency deviations and rotor-angle oscillations may become severe under weak-grid conditions with low short-circuit ratio [82].

V. MODELING OF CRITICAL COMPONENTS OF A DATA CENTER

Accurate modeling of critical data center components is essential for analyzing their power consumption behavior, dynamic response, and interaction with the electrical grid. Data centers consist of interconnected electrical, computational, and thermal subsystems, whose combined operation determines the facility's flexibility and stability support capability. Therefore, representative modeling of key components such as CPU/GPU loads, UPS, cooling systems, ESSs, and on-site renewable generation is necessary for grid integration and power system stability studies. The following section reviews the modeling considerations and characteristics of these critical components.

A. CENTRAL PROCESSING UNIT/GRAPHICS PROCESSING UNIT

Power consumption models have long been developed to evaluate energy management potential and predict power usage at both server and data center levels. In 2020, Jin et al. [83] presented a comprehensive review of 47 models that evaluate CPU power consumption in data centers. The study categorizes these models into three main types: additive models, baseline-plus-active (BA) models, and various formula-based approaches. The BA models are further divided into four categories: linear, power function, nonlinear, and polynomial forms, which are represented in (14), where $P_{s,t}$ denotes the server power consumption at time t , P_{idle} and P_{max} represent the idle and maximum power levels, respectively, $u_{s,t}$ is the server utilization, and a_1 , a_2 , b_1 , b_2 , b_3 , b_4 , and m are model coefficients defining the power-utilization relationship. Building upon this, An and Ma [84] evaluated the accuracy of BA modeling approaches and reported that polynomial-based models generally achieve better performance compared to other formulations

$$P_{s,t} = \begin{cases} P_{\text{idle}} + (P_{\text{max}} - P_{\text{idle}}) u_{s,t} \\ P_{\text{idle}} + a_1 u_{s,t}^2 \\ P_{\text{idle}} + (P_{\text{max}} - P_{\text{idle}}) (2u_{s,t} - u_{s,t}^m) \\ b_1 + b_2 u_{s,t} + b_3 u_{s,t}^2 + b_4 u_{s,t}^3 \end{cases} \quad (14)$$

Several studies have demonstrated that CPU power for fixed-function accelerators can be modeled using a register transfer level (RTL) regression-based approach [85], [86], [87]. This approach supports the development of higher resolution power models. Recent advances in machine learning have enabled more accurate power consumption modeling techniques for CPU, with predictive modeling considered as an alternative to cycle-accurate simulation [88]. In [89], advanced machine learning methods utilizing deep neural networks have shown the ability to achieve highly accurate RTL power estimation using large training datasets. More recently, Kumar et al. [90] proposed a machine-learning-based microarchitecture-level power modeling approach, using high-level activity traces and a small set of gate-level simulations. Cycle-accurate subcomponent models are hierarchically composed in the study to estimate full-CPU power, achieving less than 3.6% mean absolute error for cycle-by-cycle power of real-world applications.

GPU power consumption requires a different modeling approach compared to CPU due to their unique architectural and computational characteristics [91]. Multiple studies have been published to support this claim; for instance, a review on statistical power modeling by Chitkara [92] highlights that GPUs have distinct power dynamics due to heterogeneous computing. In 2021, Kandiah [93] developed an AccelWattch specialized GPU power model that resolves long-standing modeling challenges, specifically addressing GPUs' static and dynamic power consumption patterns. Alavani et al. [94] further validated this by developing machine learning models that capture GPU-specific power consumption relationships, achieving an impressive R^2 value of 0.9646 for the Volta architecture. Due to the rapidly evolving nature of AI technologies, Li and Li [95] classified two different workloads: training and inference. They then provided a formal mathematical model for each of the workload power consumption in (15) and (16)

$$P_{\text{train}}(t) = P_{\text{base}} + \sum_{i=1}^N \alpha_i f_i(t) + \epsilon(t) \quad (15)$$

$$P_{\text{inf}}(t) = P_{\text{idle}} + \sum_{k=1}^M P_k(t) \cdot R_k(t) \quad (16)$$

where P_{base} denotes the baseline power consumption, typically in the range of 60%–70% of the peak load. The summation term α_i represents the contribution of power variations associated with different training phases, with each coefficient reflecting the relative intensity of the corresponding phase. The function $f_i(t)$ describes the temporal profile or duration of each phase, while $\epsilon(t)$ is a stochastic term that captures minor power fluctuations arising from microbatching effects or asynchronous background tasks. For (16), P_{idle} represents the idle power draw, and the product $P_k(t) \cdot R_k(t)$ models the power demand and arrival pattern associated with a specific request type.

TABLE 5. Comparative Summary of UPS Topologies Used in Grid-Integrated Data centers [101]

UPS Topology	Key Characteristics	Advantages	Limitations	Deployment Considerations
Line-Frequency Transformer-Based	Line-frequency isolation transformers; conventional AC–DC–AC architecture.	High robustness; strong fault tolerance; inherent galvanic isolation.	Low efficiency; bulky and heavy; low power density.	High cost and space requirements; unsuitable for AI data centers.
High-Frequency Transformer-Based	High-frequency isolation with multistage conversion.	Improved power density; reduced footprint; maintained isolation.	Higher complexity; increased losses due to multiple stages.	Thermal management and efficiency critical at high power ratings.
Transformer-Less	PWM-based AC–DC–AC conversion without isolation transformers.	Very high efficiency; compact design; scalable and modular.	No galvanic isolation; increased grounding and protection challenges.	Requires strict grid-code compliance and advanced protection schemes.

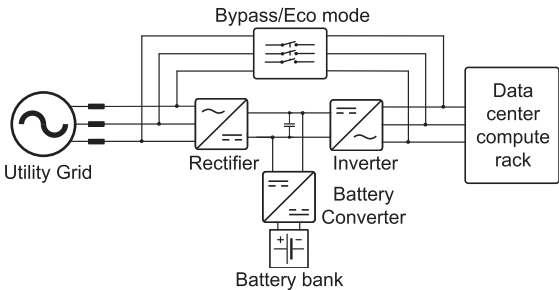


FIGURE 11. Illustrative diagram of a data center online UPS.

It is worth noting that distributed deep learning requires accurate GPU power modeling that accounts for the parallelism strategies (e.g., data, model, pipeline or fully sharded data parallelism), synchronization schemes (e.g., bulk synchronous parallel, stale-synchronous parallel asynchronous stochastic gradient descent parallel or local stochastic gradient descent) and the communication topologies (e.g., parameter server, all-reduce, and gossip) [96], [97], [98]. Thus, depending on the parallelism strategy, synchronization scheme, and communication topology, the relative contributions of computation, communication, and idle periods can vary significantly. Failure to accurately represent the idle behavior can negatively impact power modeling accuracy.

B. UNINTERRUPTIBLE POWER SUPPLY

Modern data centers are now viewed as active loads that function similarly to large inverter-based systems, making detailed modeling of UPS imperative [99]. This is essential for assessing the data center’s dynamic interaction with the grid, including power quality, load controllability, fault response, and ride-through capability under both normal and disturbed operating conditions, to ensure reliable planning, integration, and operation of power infrastructures. UPS systems can be classified into three main categories: standby (offline), line-interactive, and online (double-conversion) UPS [21]. Each type offers a distinct tradeoff when it comes to protection level, efficiency, and transfer time [100]. The online double-conversion UPS illustrated in Fig. 11 is widely used in data centers to provide the highest level of reliability, fully isolating critical loads from grid voltage and frequency

disturbances. Beyond the functional classification, UPS systems can also be distinguished based on their topologies. According to [101], power converter topologies for UPS systems can be classified into three categories based on their operating principles: line-frequency transformer-based, high-frequency transformer-based, and transformerless configurations. As summarized in Table 5, each topology presents different tradeoffs in terms of efficiency, power density, reliability, protection requirements, and deployment suitability for modern data centers.

UPS configurations differ in terms of the number of semiconductor devices used, and these differences fundamentally influence the corresponding configuration. For instance, the Vertiv Liebert EXM2 UPS [102] combines a three-phase high-frequency rectifier with a three-level T-type insulated gate bipolar transistor inverter using space vector PWM control, whereas EXL S1 UPS [103] employs a three-level neutral-point-clamped topology for both rectifier and inverter stages in large-scale applications such as AI data centers. Therefore, for EMT modeling, it is essential to identify the specific UPS converter configuration under study in order to accurately capture its distinct dynamic behavior. Once the UPS configuration is defined, detailed control-oriented models must be developed for each associated power electronic converter. Numerous UPS control modeling approaches have been reported in [100] and [101], exhibiting variations in control structure and complexity. These models typically include inner current control loops, synchronization mechanisms such as phase-locked loops (PLLs), and outer control loops for voltage, power, or DC-link regulation. Furthermore, practical implementation aspects such as current saturation and limiting strategies, filter dynamics, energy management of the ESS, ride-through control, and protection functions must be incorporated, as they play a critical role in accurately analyzing the dynamic behavior of UPS systems.

C. COOLING SYSTEMS

Modeling a data center cooling system is one of the most challenging aspects when carrying out a facility-level analysis, whether it involves air-side or liquid-side cooling. Generally, these systems have a significant impact on key performance metrics, such as PUE and water usage effectiveness [104]. Hence, modeling their power consumption

accurately is essential for effective energy management. A dynamic Modelica-based modeling framework is proposed in [105] to analyze the performance of data center cooling and control systems under both normal and emergency operating conditions. This modeling approach captures the multidomain dynamic modeling of cooling, electrical, and control systems. Furthermore, calibration against real measurement data shows that Modelica-based simulations can effectively identify operational inefficiencies and quantify significant energy-saving potentials through improved control strategies and system optimization [106]. Hence, it is particularly suitable for assessing operational behavior, identifying inefficiencies, and evaluating energy-saving potentials.

The modeling of a cooling system in the thermal liquid domain, with applications in power system studies, is proposed in [59]. The interdependence between server workload, power consumption, and thermal behavior, which enables more accurate and realistic modeling, is implemented using MATLAB [59]. The same work further demonstrates that environmental conditions have a significant impact on the resulting data center power consumption. Recently, an energy-performance modeling approach for direct liquid cooling is presented in [107] to assess the influence of supply water temperature on overall system efficiency. The developed model is validated using operational data from the HAWK supercomputer, and the results show that increasing the supply temperature improves liquid cooling energy efficiency, while also increasing heat transfer to the server room air and IT power consumption. A different study [108] develops detailed component-level mathematical models for a cold-plate liquid-cooled data center, enabling the derivation of system power consumption and chip temperature characteristics. Experimental validation confirms the high accuracy of the proposed method, supporting its applicability for optimizing operational parameters.

D. ENERGY STORAGE SYSTEMS

ESSs in data center UPSs are gradually becoming grid-interactive, supporting power system reliability rather than operating solely as backup devices. For this reason, it is essential to model these systems in a way that conforms with the battery characteristics to effectively assess the dynamic interactions between data centers and the power grid [109]. A comprehensive study of various ESS modeling techniques is presented in [110], where the approaches differ in complexity depending on application requirements, computational efficiency, modeling fidelity, and efficiency considerations. ESSs can be broadly classified into electrochemical, electrostatic, electromechanical, and electromagnetic technologies, which are typically characterized based on their power and energy density [111].

When considering long-term planning and technoeconomic analyses, bucket models [110] or energy-based and power balance models are commonly used to represent the behavior. These models typically describe the progress of the state of charge (SoC) using straightforward efficiency and capacity

constraints, which makes them computationally efficient for large-scale simulations. Such approach has been used in [112] and [113] to assess the role of ESSs in providing grid support. For dynamic and control-oriented studies, equivalent circuit models have been extensively reported in [114], [115], [116], and [117]. These models typically employ first-order and second-order Thevenin models, which are particularly popular due to their ability to capture battery voltage dynamics, internal losses, and transient behavior under rapidly varying loads. A more detailed model with a high degree of accuracy relies on electrochemical models, which describe ion transport, diffusion, and reaction kinetics within the battery cells. A comprehensive description of the underlying equations and the methodology associated with this modeling approach is provided in [118], [119], and [120]. In addition, the effective use of this modeling approach requires a clear understanding of the ESS's architecture and operating behavior [110].

E. OTHER SOURCES (WIND AND SOLAR PLANTS)

Wind and solar plants are increasingly integrated on-site or locally with data centers to meet decarbonization and cost objectives [121]. In such configurations, local generation and storage must closely track highly dynamic data center loads, fundamentally changing the power balance and operational requirements. Unlike centralized renewables, these resources operate in a tightly coupled environment where inverter-based generation, energy storage, and load dynamics interact over fast time scales, necessitating accurate converter-based modeling. Both phasor-domain and EMT models are used, with EMT becoming critical under weak-grid and islanded conditions [72]. Review studies show that hybrid renewable architectures can enhance reliability and reduce dependence on diesel backup, but they also introduce protection challenges such as bidirectional power flows, low fault currents, and complex coordination with UPS systems [122], [123]. With regard to this, robust modeling must account not only for energy production but also for control dynamics and adaptive inverter-aware protection schemes to ensure resilient operation of AI data center loads.

VI. INTERCONNECTION REQUIREMENTS

Over the years, grid codes have primarily concentrated on generation assets, requiring capabilities such as FRT, reactive power support, voltage control, and frequency regulation [125]. Large electrical loads, including traditional data centers, were generally treated as passive consumers and, therefore, subjected to comparatively limited operational requirements. However, large-scale AI data centers operate as converter-dominated infrastructures with highly dynamic demand characteristics, capable of influencing both local and wide-area grid behavior. Due to this outcome, several system operators and regulatory bodies are beginning to recognize that large-scale data centers should satisfy interconnection and operational requirements similar to those traditionally imposed on generation facilities [125].

TABLE 6. Proposed Grid-Code Requirements for Grid-Integrated Data Centers Across Selected System Operators

System Operator	Proposed Grid-Code Requirements	Required Capabilities	Defined Thresholds	Applicable to	Motivation and Technical Rationale
ERCOT (Texas) [124]	Limits on data center load ramp rates; mandatory frequency-responsive demand; fast load shedding; coordination of UPS and on-site ESS for inertial and primary frequency support.	Provision for remote load curtailment or disconnection, capability to island using backup resources, comprehensive interconnection modeling, and cost-sharing mechanisms	On-site generation with ERCOT authority to curtail load or switch to backup supply	Large data centers (≥ 75 MW).	High penetration of inverter-based resources and limited interconnection capacity increase sensitivity to rapid load variations.
AESO (Alberta) [13]	Maximum load ramping constraint; controlled reconnection after outages; voltage and frequency ride-through capability.	Ramp rate control logic; phase jump; ride-through and reactive power capability; project modeling	Ramp rate 10 MW/min; power factor ± 0.95 ; voltage control 0.90–1.10 p.u.; RoCoF ≤ 5 Hz/s; phase jump $\leq 25^\circ$; V/Hz 1.1 p.u. (45 s); PF tolerance $\pm 1.75\%$; recovery time 1–10 s; oscillation ± 160 kW; variability < 16 kW/100 ms	Transmission connected data centers (≥ 75 MW).	Prevention of transient instability caused by sudden load loss or reconnection.
Fingrid Oyj (Finland) [125]	Mandatory participation in Frequency Containment Reserve (FCR); active power–frequency control with defined response times.	LVRT/HVRT, phase-jump immunity, multi-fault ride-through, controlled post-fault ramp, dynamic voltage/reactive behavior, TSO signaling	RoCoF ≤ 2 Hz/s; $\pm 30^\circ$ phase jump without tripping; 10×100 ms LVRT events within 90 s; active current limited to < 0.9 p.u.; blocking for $V < 0.5$ p.u.; OVRT defined per kV base; staged recovery	Demand sites data center > 30 MW	Support of system stability in a low-inertia Nordic power system.
CRU (Ireland) [126]	Controlled reconnection following outages; limits on demand recovery; enhanced real-time monitoring and reporting.	Generation/storage must match load on-site/locally, can participate in the demand response, requires annual renewables and emissions reporting, and is subject to location constraints.	Dispatchable generation/storage meets demand; constrained-zone operators face stricter oversight	New data centers (all).	Low system inertia and high sensitivity to large step changes in demand.

Recent developments by the ERCOT, AESO, Fingrid Oyj, and the CRU Dublin reflect this transition toward more structured regulatory monitoring and enforcement for large-scale data center interconnections. Table 6 summarizes the emerging interconnection requirements, operational concerns, and evolving compliance expectations introduced by these system operators for large-scale facilities. This section discusses four key interconnection requirements identified across these regulatory frameworks, distinguishing between requirements that are currently mandatory, those regarded as recommended practices, and those that are still evolving toward formal compliance obligations.

A. POWER QUALITY REQUIREMENTS

Power quality compliance at the PCC has become one of the most important interconnection considerations for converter-dominated large-scale data centers. The use of UPS systems, switched-mode power supplies, battery converters, and variable-speed cooling drives introduces harmonic distortion, voltage flicker, interharmonics, and current imbalance [127]. IEEE 519 remains the dominant framework governing harmonic emission limits at the PCC. The standard defines

allowable current and voltage distortion levels based on PCC voltage level and the ratio between available short-circuit current and facility load current [128]. Although IEEE 519 is formally published as a recommended practice, its harmonic limits are widely enforced through utility interconnection agreements and tariff requirements, making compliance effectively mandatory for large-scale AI data centers connected to the grid.

A report by the Energy Systems Integration Group [129] has identified that fast dynamic behavior of large loads can lead to subsynchronous interactions that conventional static load assumptions do not adequately represent. A significant issue is that the high-frequency cycling behavior associated with converter controls may contribute to subsynchronous control interactions, subsynchronous torsional interactions, and resonance phenomena, particularly under weak-grid conditions. Consequently, utilities and system operators now require large-scale data centers to comply with harmonic emission standards such as IEEE 519 [130]. More specifically, regional approaches vary considerably depending on local grid characteristics and system strength. The AESO mandates on detailed harmonic and EMT studies for

large-scale converter-dominated weak portions of the Alberta network [13], while Nordic system operators emphasize on harmonic monitoring and voltage quality assessment due to growing concentrations of digitally intensive loads [125].

B. LVRT/FRT AND GRID SUPPORT

Ride-through capability represents one of the most rapidly evolving areas of large-load interconnection requirements, highlighting a significant gap between existing mandatory standards and the performance expectations emerging from grid operators. During transmission disturbances, severe voltage depressions at the POI may trigger rapid disconnection of UPS systems, server power supplies, and converter-based infrastructure designed to protect sensitive IT equipment [19]. The simultaneous loss of several large facilities can abruptly remove hundreds of megawatts of demand from the grid, resulting in frequency excursions, uneven voltage recovery, transient power imbalance, and reduced operator visibility for maintaining system stability. This requires grid-integrated data centers to remain connected during temporary voltage and frequency disturbances through defined ride-through capabilities [129]. These requirements are becoming mandatory in weak grids and low-inertia systems, where sudden large-load tripping can significantly amplify instability risks [13].

Several jurisdictions have now introduced detailed ride-through requirements for large transmission-connected loads and data centers. The Southwest Power Pool, through its high-impact large load framework, specifies mandatory voltage ride-through envelopes, controlled active power recovery requirements, and restrictions on constant-power converter operation during voltage disturbances to prevent excessive current amplification under LV conditions [131]. The AESO also proposed voltage tolerance, frequency withstand, and phase-angle disturbance requirements for grid-integrated data centers to mitigate instability risks associated with sudden load disconnection [13]. Ireland's increasing data center penetration has prompted regulators to emphasize coordinated reconnection procedures and operational flexibility following network disturbances [126].

C. DYNAMIC MODELING AND OPERATIONAL VISIBILITY

The increasing complexity of data centers have created growing demand for accurate dynamic models capable of representing fast load transients, converter interactions, backup generation behavior, and coordinated control systems [59]. With AI data centers exhibiting extremely fast demand ramps associated with computational workload scheduling, GPU cluster activation, and cooling system responses, high-fidelity modeling approaches are necessary for accurate representation of their grid interactions [132]. As a result, several system operators and utilities now require, and in some jurisdictions mandate, the provision of real-time telemetry and accurate dynamic models as part of large-load interconnection requirements [13], [131]. In the ERCOT, growing concerns regarding large coincident load ramps and operational uncertainty have motivated increased emphasis on operational telemetry,

controllability, and real-time visibility for large transmission-connected loads [124]. Fingrid further require validated EMT and phasor-domain models benchmarked against measured operational data for compliance verification [125].

Operational visibility requirements are also expanding beyond SCADA monitoring. Advanced telemetry, high-speed measurements, real-time operational coordination, and demand forecasting are increasingly viewed as essential for maintaining grid situational awareness under high penetrations of flexible and rapidly varying loads [133]. Most regional transmission organizations require large transmission-connected customers, typically those exceeding 20 MW, to supply SCADA-based telemetry data, including measurements of active power, reactive power, and voltage, to system operators for real-time monitoring and operational coordination [134]. While behind-the-meter visibility, such as the operating status of UPS, generator dispatch, and battery SoC, is not yet universally mandatory, organizations like the Department of Energy and the North American Electric Reliability Corporation recommend enhanced monitoring and telemetry to improve grid visibility, reliability, and operational awareness. These developments indicate that future interconnection frameworks will likely require data centers not only to demonstrate compliance during planning studies but also to maintain continuous operational transparency throughout their life cycle.

D. INTERCONNECTION STUDY PROCESS

Interconnection approval for large-scale grid-integrated data centers typically requires comprehensive technical studies to assess potential impacts on transmission system reliability, stability, protection coordination, and infrastructure adequacy. Unlike smaller commercial loads, these facilities can affect regional load forecasting, transmission expansion planning, voltage stability margins, and operational reserve requirements. Accordingly, utilities and system operators commonly recommend staged interconnection assessments that may include power flow analysis, short-circuit studies, harmonic analysis, transient stability simulations, EMT studies, and protection coordination evaluations [13], [129], [131]. The scope of required studies generally depends on grid-specific thresholds such as connection capacity, voltage level, geographical location, and proximity to constrained network regions [133]. Fig. 12 illustrates the relationship between simulation domains and the characteristic time scales associated with grid interconnection studies. The figure also highlights the studies required at different stages of the connection process. EMT studies are primarily required for fast electromagnetic phenomena occurring within microsecond-to-second time scales, while phasor-domain simulations are generally applied to slower electromechanical phenomena [135].

VII. GRID INTEGRATION CHALLENGES

As data centers continue to evolve into active participants in modern power systems, grid integration challenges have become a critical focus for ongoing research. The literature

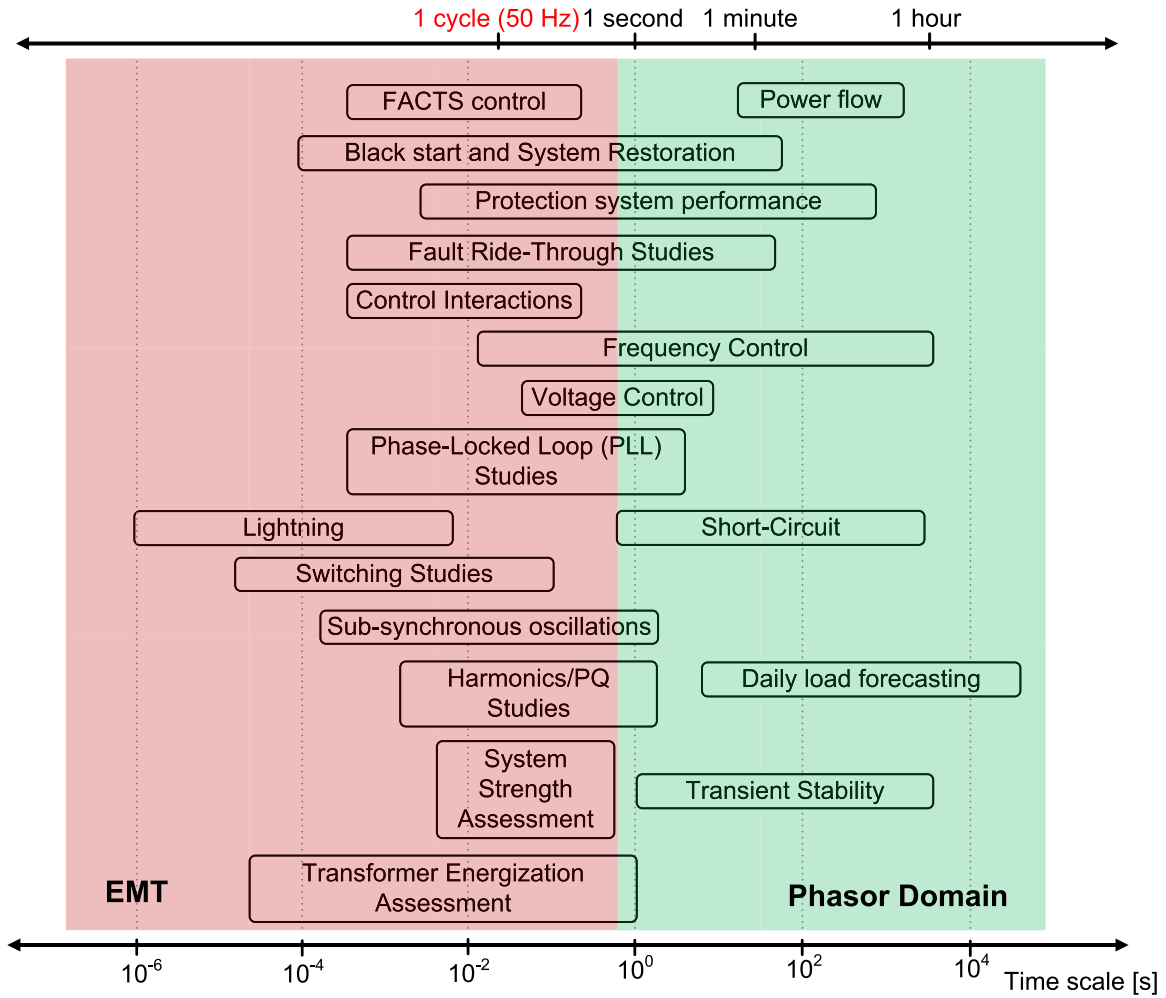


FIGURE 12. Time scales of power system dynamics, control actions, and required studies during different stages of data center grid connection.

highlights a range of technical issues, including power stabilization during high power ramps, harmonic interactions, system stability, voltage quality, and FRT capability. Further discussions of these challenges are provided in the following subsections.

A. POWER STABILIZATION AND HIGH POWER RAMPS

Computing demand can create extreme power dynamics that can lead to undesirable high-magnitude transient peaks in electricity consumption within subsecond to minute timescales [1]. A study by Jimenez-Ruiz and Milano [62] on the all-island Irish transmission system reveals that uncontrolled data center reconnection timing can lead to grid instability due to a sudden surge in electrical power demand. Motivated by such risks, the AESO have introduced explicit ramping constraints to limit load ramping to a maximum rate of 10 MW/min [13]. Another study [95] shows that large-scale AI workloads impose unique demand on power conversion chains and that synchronized GPU operations can induce facility-wide power transients that strongly influence the dynamic behavior. Building on [62] and [95],

Peivandizadeh [136] proposed a new stability criteria for gigawatt-scale ramps, where the critical clearing times for protection systems should be reduced from 150 to 83 ms.

In 2025, researchers from Microsoft, OpenAI, and NVIDIA examined the challenges associated with power swings caused by the high variability in power consumption during training workloads [30]. They subsequently proposed three classes of power stabilization techniques to mitigate these swings, including: 1) software-based approaches that inject controlled workloads to smooth power transitions; 2) GPU-level firmware mechanisms that enforce ramping constraints and power floors; and 3) rack-level energy storage solutions that absorb and release power as needed. In addition, Huntington and Tu [27] proposed the use of BESSs, and Siemens Energy [137] introduced the deployment E-STATCOM, both installed adjacent to the grid interconnection to smooth transient power swings. Meanwhile, Ko et al. [138] presents a hybrid coordinated control of colocated BESS and E-STATCOM. Collectively, these existing solutions can be categorized into software-defined approaches operating within the computing stack and grid-compatible solutions that leverage power

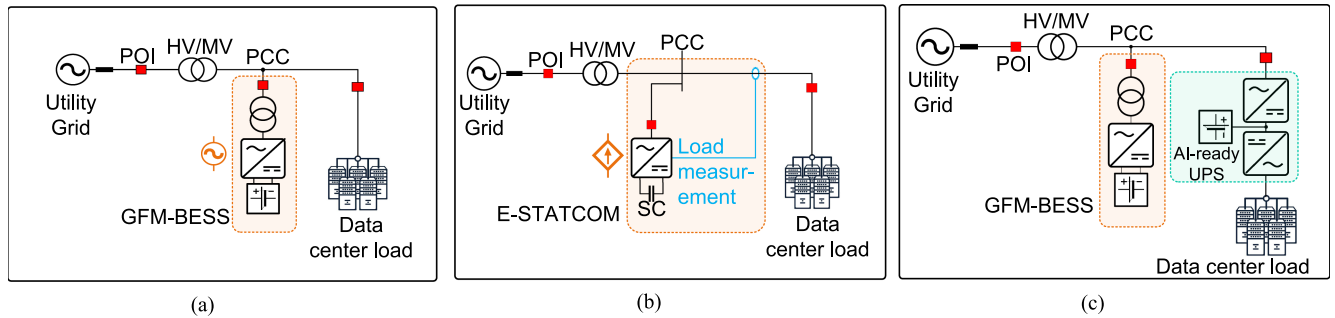


FIGURE 13. Grid-integrated data center support architectures for power system stability. (a) GFM-BESS. (b) E-STATCOM. (c) Hybrid approach.

TABLE 7. Benefits and Challenges of Storage and Control Solutions for AI Data Centers

Solution	Benefits	Challenges
E-STATCOM	Provides very fast dynamic response, mitigates voltage fluctuations, smooths rapid load variations, and improves power quality at the PCC.	Primarily designed for local power quality support; limited capability for wider grid-service participation and typically lacks inherent GFM functionality.
Grid-Forming BESS	Enhances load balancing and operational flexibility while contributing to voltage regulation, frequency support, and overall grid stability.	Requires sophisticated supervisory control and coordination; additional interface components such as line reactors may be necessary, and transient flicker issues can still arise under aggressive operating conditions.
Hybrid approach	Combines fast compensation with energy buffering, enabling improved ramp-rate control, gridsupport services, and enhanced system resilience.	Involves greater implementation cost, larger installation footprint, and increased operational complexity due to multi-device coordination.

electronic and energy storage technologies at the grid and load interface. Fig. 13 illustrates representative grid support architectures for grid-integrated data centers aimed at improving power system stability and operational flexibility. Specifically, the GFM-BESS configuration for voltage and frequency support, is an E-STATCOM-based architecture for reactive power compensation and voltage regulation and is a hybrid GFM-BESS with AI-ready UPS architecture that enhances system resilience, controllability, and disturbance ride-through capability through coordinated converter-based support. Table 7 further summarizes the corresponding advantages, operational benefits, and implementation challenges associated with these grid-compatible support architectures.

B. HARMONICS

Large-scale data centers loads are typically characterized as nonlinear loads that introduce significant harmonic distortion into the grid, resulting in power quality degradation. Therefore, harmonic distortions can accumulate in large facilities housing thousands of server racks, spreading through the local distribution network and posing hazards to nearby industrial and residential systems, including equipment malfunctions, accelerated degradation, overheating, and even potential electrical fires [1]. An analysis reported by Bloomberg in 2024 highlights that over three-quarters of highly distorted power quality measurements in the United States occur within 50 miles of major data center activity [139].

An extensive study conducted by North American Electric Reliability Corporation [82] also points out that large-scale

data centers in MW sizes can stimulate torsional modes in nearby generating stations due to interharmonics, resulting in damage to turbine shafts. Zubi and Hfouda [140] investigated the power quality of a typical data center, revealing that heavy-loaded nonlinear loads leads to poor power quality at the PCC, with several scenarios approaching or exceeding the limits specified in IEEE 519 standard. That is, as more data centers connect to the grid, harmonics will increase further, thereby pushing the total harmonic distortion beyond acceptable limits [141]. This situation poses significant compliance challenges for utility operators.

C. STABILITY

Several stability challenges are emerging on the grid due to the increasing integration of AI workload data centers. Stability is evaluated based on multiple criteria, including frequency, voltage, rotor angle, converter-driven, and resonance [142]. Highly dynamic data center operations that supply large nonlinear loads can impact the aforementioned criteria, leading to grid instability [143]. Such stability issues can increase stress on grid assets, accelerate equipment degradation, and ultimately compromise overall grid reliability. A practical example of frequency stability challenges occurred in all-island Irish transmission system [144]. Where a fault-induced event led to a sudden reduction of 204 MW in data center demand. This disturbance caused a rate of change of frequency (RoCoF) of 0.12 Hz/s, reaching a peak frequency of 50.22 Hz.

These challenges are further complicated by the geographic clustering of data centers, which creates large and highly

dynamic power demand [145]. Subsynchronous resonance has emerged as a major challenge in situations like this [77]. A study, motivated by a real-world example [146], investigated the high-frequency undamped oscillations that appeared, initiated by a periodic voltage disturbance in a data center-dense region within Dominion Energy's grid. The ERCOT, WECC, and PJM Interconnection also reported situations where large computing loads tripped and disconnected from the grid during transmission disturbance [1], [147], [148]. Kwon et al. [149] investigate how large-scale data centers dynamically interact with the grid and have shown that sudden load surges cause significant frequency deviations, persistent oscillations may lead to system collapse, and gradual load ramps tend to trigger slower oscillatory behaviors. Research on wide area power systems with integrated large-scale AI workloads indicates that oscillation modes are significantly influenced by load fluctuation frequency and the size of data centers [132].

D. VOLTAGE FLICKER

Another power quality issue arising from the integration of data centers into the grid is voltage flicker. This issue is typically caused by voltage fluctuations that result from dynamic changes in electrical load currents interacting with the system impedance [150]. A field measurement investigation conducted at the PCC between the grid and two large data centers (20 and 32 MW) by Nassif et al. [127] revealed that multiple sources, including large HVAC systems, massive computer loads, sudden power demand changes, and nonlinear power electronic equipment, tend to generate voltage flickers. This is further confirmed by Ahmed et al. [151], which adds that internal power condition systems in data centers are prone to faults that can create voltage flickers, which propagate through the electrical system. The recent report from [82] highlighted that large loads, such as cryptocurrency miners and AI data centers, can cause voltage flickers in the grid. This occurs due to the variable load profiles associated with AI workloads, which may lead to voltage flicker levels exceeding the acceptable limits outlined in IEC 61000-3-3 [152].

E. FAULT RIDE-THROUGH/LOW-VOLTAGE RIDE-THROUGH

As power electronic converters increasingly dominate the power grid, FRT capability has become essential for maintaining grid stability. This capability allows generation assets and equipment to remain connected during faults, such as voltage sags [153]. When a system can sustain and support grid voltage during these fault induced voltage sags, it is referred to as LVRT [154]. Traditionally, LVRT requirements have been associated with generators, while loads have generally been considered passive. However, with AI data centers' power demands reaching hundreds of megawatts, these loads are now seen as active participants in the grid. Thus, FRT capabilities of such large loads present significant challenges for grid stability. These challenges are no longer theoretical, as recent grid disturbances have demonstrated that inadequate FRT capability of large-scale AI data center loads can directly translate into widespread system-level stability impacts.

A practical example was recorded on 2024, when a permanent fault occurred on a 340-kV transmission line within the Eastern Interconnection in the United States [155]. The disturbance triggered a sequence of six short-duration voltage violations over an interval of approximately 82 s, with individual events lasting between 42 and 66 ms and reaching magnitudes of 0.24–0.40 p.u. The resulting under voltage conditions activated demand-side protection mechanisms, leading to the disconnection of nearly 1.5 GW of voltage-sensitive load [155]. Postevent investigations determined that this entire curtailed load originated from data center facilities. Following the tripping events, the system voltage increased to approximately 1.07 p.u.

In response to the issue highlighted above, system operators, particularly those that have not previously imposed ride-through requirements on large-scale data centers, are now actively evaluating the introduction of such requirements. For example, AESO technical interconnection requirements state that transmission-connected data centers must be designed to withstand voltage disturbances [13]. In addition, NERC has formed a dedicated task force to examine the system-level impacts associated with large load interconnections [156]. Similarly, the ERCOT has established a large-load working group and is currently consulting on the implementation of FRT requirements for loads with capacities exceeding 75 MW [124]. In Europe, EirGrid and FinGrid have proposed updated FRT standards for large energy users [125]. In Australia, the Australian Energy Market Operator proposed a rule change to be finalized in 2026 regarding the access standards for large loads, such as large inverter-based loads [99].

F. CYBERSECURITY AND DATA SECURITY

Grid-integrated data centers concentrate vast quantities of sensitive, proprietary, and regulated information while simultaneously interacting with critical power system infrastructure, making data security and cybersecurity one of the most consequential operational and grid-reliability challenges in modern digital energy systems. The progressive convergence of IT and operational technology within these facilities has substantially expanded the cyberattack surface. Systems that were traditionally isolated, including SCADA controllers, building management systems, UPS management buses, cooling controllers, energy management systems, and data center infrastructure management platforms, are now connected to IP-based and cloud-accessible environments [157]. This convergence exposes not only digital assets but also electrical and thermal control infrastructures to adversarial threats that can affect both facility operations and power system stability.

A survey reported that nearly 90% of organizations operating connected IT infrastructures had experienced cybersecurity breaches within their ICS/SCADA environments [158]. In grid-integrated data centers, such attacks may extend beyond data compromise to directly influence grid operation through manipulation of electrical demand, disruption of demand response participation, or destabilization of distributed energy resources. Since large-scale AI facilities can consume

TABLE 8. Summary of Opportunities for Grid Support From AI Data Centers and Their Impact

Opportunity	Limitation Addressed	Potential Impact	Methods
UPS-Based	Rapid power fluctuations and lack of immediate response resources	Millisecond-scale response for frequency containment and transient stabilization	Controlled charge/discharge of AI-ready UPS while maintaining state-of-charge margin
Cooling Control	Limited short-term demand flexibility in conventional loads	Secondary frequency support and peak demand reduction	Dynamic adjustment of cooling setpoints, leveraging thermal inertia, predictive cooling control
Grid-Forming BESS	Low inertia, renewable intermittency, and voltage instability	Synthetic inertia, fast frequency response, voltage support	Deployment of grid-forming inverters at PCC with configurable active/reactive power control
On-Prem Generator	Peak demand stress and supply shortages	Long-duration backup support and reduced grid dependency	Dispatchable operation of generators or fuel cells during peak or contingency events
E-STATCOM	Voltage instability and fast load transients from AI hardware	Sub-cycle voltage regulation and power transient mitigation	Integration of supercapacitor-based STATCOM systems for rapid active/reactive compensation
Workload Scheduling	Rigid demand profiles and limited demand-side participation	Improved load balancing and renewable energy utilization	Temporal shifting of non-urgent workloads using intelligent schedulers
Workload Migration	Regional grid congestion and renewable variability	Geographical load balancing	Relocation of flexible AI workloads across geographically distributed data centers
Clock Rate Modulation	Lack of fine-grained real-time load control	Fast demand shaping and primary frequency support	Dynamic adjustment of CPU/GPU voltage and frequency within SLA constraints

hundreds of megawatts, malicious alteration of cooling systems, workload scheduling, UPS dispatch, or battery coordination may trigger rapid load fluctuations, voltage instability, frequency deviations, and local network congestion within transmission and distribution systems.

VIII. OPPORTUNITIES

Modern data centers are evolving into controllable grid-interactive assets. Their integrated electrical, thermal, and computational systems enable coordinated multitimescale flexibility, allowing effective participation in grid support services. These positions them as enablers of future low-inertia and renewable-dominated power systems. The following section discusses these flexibility services and enabling technologies in detail, demonstrating how AI data centers can provide fast, reliable, and scalable grid support.

A. FREQUENCY CONTROL SERVICES

Data centers can provide additional services to help stabilize the grid, especially now that they are being redefined as grid-interactive assets [125]. Numerous studies and real-world applications have demonstrated that when data centers are properly coordinated, they can effectively contribute to ancillary services and system balancing [159], [160], [161]. These facilities are capable of adjusting active power consumption within subsecond to minute timescales, due to their fast-acting power electronic interfaces, flexible workload scheduling, and integrated ESSs. This makes them well suited for frequency control services.

Recent studies demonstrate that data centers can effectively support power system frequency regulation through coordinated load and energy storage control. A coupled

electrical–thermal modeling study of a 10-MW data center showed that participation in demand response, via IT load reduction and cooling setpoint adjustment, improved grid frequency from 59.58 to 59.60 Hz following a 25% load disturbance on the grid [59]. By leveraging real-time workload modulation and UPS coordination, another study reported an increase in frequency nadir from 59.66 to 59.85 Hz and a 34% reduction in recovery time under a 200-MW generation loss [162]. A cooperative control of delay-tolerant workloads and backup power supplies across 350 MW of data center capacity was shown to contain frequency deviation above 49.6 Hz and limit RoCoF to below 730 mHz/s in ultra-low-inertia systems operating at 75% system nonsynchronous penetration [163]. In addition, a data-driven co-optimization framework demonstrated that higher penetration of flexible data center loads consistently improves system performance and enhances renewable integration [164].

B. FLEXIBILITY SERVICES

They have various internal assets and operational control strategies that can be utilized to provide support for the grid while still meeting SLAs for core computational tasks [165]. After reviewing the frequency control services offered by data centers, it is important to identify the different sources of flexibility for these services. Each source exhibits distinct operational characteristics that influence its suitability for delivering ancillary services. The following subsection discusses the flexibility assets, response time, and their applications.

1) UPS-BASED

A UPS is traditionally designed and viewed as a reliability device, used to secure critical loads in the data center during

power interruptions, with real-world demonstrations showing that they can respond quickly during frequency-containment disturbances [166]. Today, AI data centers are equipped with AI-ready UPS systems [167]. These systems are large GPU clusters that experience rapid power fluctuations when accelerators synchronize during training or inference. These events occur faster than most control systems can respond, making the UPS the first line of stabilization because it is always online and electrically connected to the internal bus. In this context, the UPS serves as a transient shock absorber rather than just a backup energy source. It can inject or absorb power within milliseconds, effectively providing inertial and primary frequency response for ancillary services markets. From the grid's perspective, during these conditions, the data center behaves like an active load. The key limitation is availability rather than power rating. The stored energy in a UPS primarily exists for ride-through capability, which means only a fraction of its capacity can be used for grid services while still maintaining contingency reserves. Thus, the participation of UPS systems as a flexible resource in the future must be carefully managed within SoC margins and closely coordinated with the facility's energy management system [162].

2) COOLING CONTROL

Cooling systems in data centers account for a significant portion of total electrical demand, following IT load demand, making them an important source of flexibility. Unlike core IT loads, cooling demand has inherent thermal inertia that can be leveraged by temporarily adjusting temperature setpoints [59]. Operational strategies include precooling before anticipated peak periods or temporarily relaxing cooling requirements when grid support is needed [168]. This approach can be applied in high-power-density data centers, where the thermal inertia of liquid cooling loops, air volumes, and server room structures acts as short-term energy buffers. These buffers can absorb and release energy over time, helping to smooth out cluster-level power ramps caused by GPU utilization. Cooling adjustments respond slowly due to thermal lags and the slow propagation of heat. However, they can significantly contribute to secondary frequency regulation for events that last minutes. Nevertheless, thermal margins can be limited by the operational envelopes and standards of the hardware; therefore, it is essential to manage flexibility carefully to prevent accelerated degradation of components.

3) GFM-BESS

As mentioned in Section III, modern data centers now deploy dedicated BESS as grid-interactive assets located at the PCC. This approach is already used by Google and Microsoft, where lithium-ion BESS are integrated into their data center infrastructures, primarily to enhance operational resilience and support renewable energy integration [169], [170]. When operated with GFM control, these BESS units emulate the dynamic behavior of synchronous generators by actively regulating terminal voltage and system frequency, while also delivering synthetic inertia during fast grid transients [143].

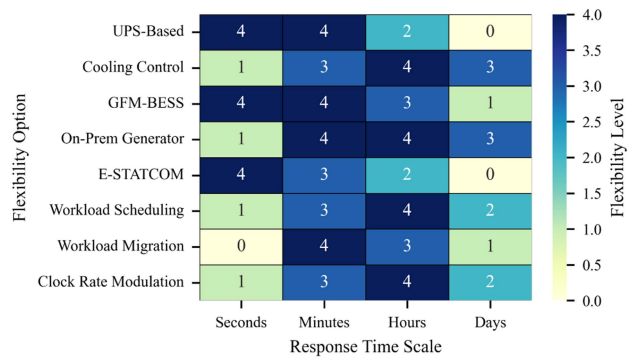


FIGURE 14. Data center flexibility options across time scales.

This capability has been demonstrated in hyperscale data centers, where GFM Tesla Megapack systems help stabilize AI load variability and improve voltage resilience during grid disturbances [171]. Although energy-limited, these assets can provide controllable ramp rates and configurable participation for fast frequency response, making them the primary electrical buffer in AI data centers.

4) ON-PREM GENERATOR

On-premise generators provide a fundamentally different form of flexibility function compared to electronic or computational controls, most commonly using diesel, natural gas, or hydrogen-based technologies [172]. Due to environmental concerns associated with diesel generators, fuel cells have emerged as a cleaner and reliable alternative for primary and backup power in data centers, a feasibility demonstrated by a 3-MW PEM fuel cell installation in Latham, United States [173]. These assets can be viewed as deep-reserve layer of data center flexibility significant for grid support, particularly by reducing the reliance on the grid during peak demand times. However, their relatively slow ramp-up rates and regulatory restrictions regarding emissions and continuous operation limit their effectiveness for rapid frequency regulation. Instead, they are better suited for secondary and tertiary system frequency regulation, where duration and predictability are more critical than response speed [160]. Rather than reshaping demand, they temporarily replace grid supply.

5) E-STATCOM

E-STATCOMs are power electronic assets that enhance grid flexibility by integrating supercapacitors in parallel with loads to provide fast active power compensation during load swings and reactive power support for voltage stability [137], [174]. AI data centers experience rapid power fluctuations and voltage disturbances due to the large power electronic converters utilized in GPU power supplies, especially during inference and training of AI models. Unlike GFM-BESS, which effectively manages energy imbalances over longer periods, E-STATCOMs offer a fast-response solution to mitigate these power and voltage transients and can be deployed at the PCC within the data center. However, this capability comes at the cost of limited energy capacity in supercapacitor-based

storage. Their response times are typically in the subcycle to millisecond range, making them particularly effective for inertial response, fast frequency response, and voltage regulation in weak or low-inertia grids [175]. Therefore, they occupy the fastest layer of the flexibility stack, enabling the facility to host extremely dense compute hardware without compromising grid stability, while allowing slower flexibility mechanisms to respond subsequently in grid-interactive data centers.

6) WORKLOAD SCHEDULING

Data center operations involve a variety of workloads that differ in terms of urgency and computational flexibility [176]. These workloads can be broadly categorized into urgent and nonurgent tasks, based on their sensitivity to latency and execution requirements. Nonurgent tasks, such as data backups, large file transfers, and the training of new AI models, do not require immediate execution. Because of this, they can be scheduled flexibly to help support the system during periods of high demand. In 2023, Google demonstrated a workload-shifting approach that delays nonurgent tasks to support grid demand response [177]. Studies indicate that 30%–50% of current data center workloads are deferrable, a share expected to grow with the expansion of AI data centers [178]. This strategy offers flexibility on minute-to-hour timescales. While SLAs and latency requirements constrain workload scheduling, the flexibility is still substantial because modern data centers host a heterogeneous mix of interactive and batch processes. As AI workloads expand, the proportion of deadline-based computation is likely to grow, increasing the temporal buffering capability of data centers. Thus, workload scheduling should be viewed as the core load-shaping resource, forming the middle layer between fast but shallow clock rate modulation control and slower spatial mechanisms such as work load migration.

7) WORKLOAD MIGRATION

Workload migration introduces spatial flexibility into data center operation by geographically relocating computation rather than curtailing load, thereby exploiting regional differences in electricity prices, policy incentives, and renewable energy availability [179]. This capability has been shown to support grid balancing for both data centers and system operators through cost-effective coordination strategies [180]. AI workloads are becoming increasingly portable across geographically distributed data centers. Moving training tasks between regions does not change how much energy is used, but determines where the grid must supply it. However, the flexibility is constrained by data gravity and communication latency. Large datasets and network bandwidth can impose practical limits on how frequently workloads can move and how much load can be relocated. Consequently, migration operates naturally on minute-to-hour horizons fast enough to follow renewable ramps and market intervals, but too slow for primary control services [172].

8) CLOCK RATE MODULATION

Clock rate modulation, typically implemented through dynamic voltage and frequency scaling (DVFS) at the server or processor level, provides a fine-grained control lever for adjusting the power draw of compute resources [181]. By leveraging this approach on CPU and GPU cores, data centers can significantly reduce power consumption with minimal impact on computational performance, enabling effective and responsive power flexibility. For example, during AI training, underclocking an NVIDIA A100 GPU running a Vienna Ab initio simulation package (VASP) workload achieved approximately 50% power reduction with only 10% performance loss [182]. The main advantage of this approach is its proportional and reversible behavior, which allows demand shaping in small increments, with performance recovered immediately the constraint is released. This provides second response capabilities, making it suitable for primary frequency regulation [163]. The response time, measured in seconds, aligns well with the time constants of grid control, enabling DVFS to operate more like a controllable load than a delayed computational task. However, it is essential to note that DVFS should not be seen as a deep flexibility resource. Its effectiveness is limited by performance guarantees and SLAs, which cap the achievable curtailment. In practice, DVFS provides high-quality but low-volume flexibility. Consequently, it is best deployed as a regulation-layer control, complementing slower but larger mechanisms such as workload scheduling or migration.

Table 8 summarizes the key opportunities in grid-integrated AI data centers, highlighting the challenges addressed, potential impacts, and enabling methods, while Fig. 14 illustrates the available data center flexibility options across multiple time scales.

IX. CONCLUSION AND FUTURE NEEDS

Recent reports indicate that data centers are experiencing rapid growth and becoming increasingly integrated into the electrical grid, which presents new challenges. Choosing the right modeling approach for data centers is imperative to accurately represent their interaction with the grid under various operating conditions. This article presents a detailed review of data center architectures, modeling techniques, interconnection requirements, as well as the challenges and opportunities associated with grid integration. In addition, detailed modeling considerations for critical components have been discussed to support accurate representation of dynamic load behavior and grid interaction across multiple time scales.

Looking forward, future research must prioritize hybrid and high-fidelity modeling frameworks that balance computational efficiency with dynamic accuracy, particularly for large-scale AI data centers. There is also a growing need for standardized grid-code frameworks tailored to large flexible loads, enhancing coordination between data center energy management systems and system operators, as well as integrating GFM assets and machine-learning-based predictive

control strategies to effectively utilize multitimescale flexibility for improved grid stability and reliable participation in ancillary services. Addressing these needs will be essential for enabling AI data centers to operate not merely as large consumers of electricity, but as resilient, intelligent, and grid-supportive assets in low-inertia renewable-dominated power systems.

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